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Unsupervised hybrid attribute selection based on variable precision neighborhood rough sets

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ABSTRACT

Unsupervised attribute selection is an important data preprocessing method in the unsupervised knowledge discovery process, which can reduce the dimensionality of the data by removing irrelevant and redundant attributes. It aims to identify an optimal subset of attributes that preserves the original data distribution as closely as possible. Most of the existing unsupervised attribute selection methods are mainly focus on a single attribute type (nominal or numerical). They are not able to handle uncertain hybrid data effectively. In real life, data usually exists in uncertain hybrid forms. As an extended model of rough sets, neighborhood rough sets can effectively handle uncertain hybrid data. It has been widely used for supervised hybrid attribute selection, while unsupervised hybrid attribute selection needs to be further explored. Based on these observations, this paper proposes an unsupervised hybrid attribute selection strategy considering conditional attributes. First, a variable precision neighborhood rough set model is constructed for a decision-free information system. Compared with the neighborhood rough set model, this model has better noise resistance and fault tolerance. Next, the variable precision neighborhood dependency and the increase rate of the average variable precision neighborhood correct classification rate of the attribute are defined based on all single-condition attribute subsets, based on this, the neighborhood attribute quality degree is defined, which can serve as the attribute evaluation metric. Further, an unsupervised hybrid attribute selection algorithm is designed based on neighborhood attribute quality degree. Finally, the proposed algorithm is compared with existing unsupervised attribute selection algorithms. Experimental results show that the algorithm is effective and superior in handling uncertain hybrid data. The code is publicly available online at https://github.com/Scmz-YiLi/UAS_NAQD.

1. Introduction

With the coming of the data era, massive and high-dimensional data can be found everywhere in various industries. The high-dimensional data often contains a large number of irrelevant and redundant attributes. If these high-dimensional data are utilized directly in practical applications, not only is the computational overhead high, but also the performance of algorithms will be decreased (Li et al., 2021). Dimensionality reduction methods are usually used to preprocess high-dimensional data (Yin et al., 2024; Yuan et al., 2021a). Currently, the common dimensionality reduction methods are mainly divided into two categories, which are attribute extraction (Nie et al., 2019a,b) and attribute selection (Chen et al., 2022a; Liao et al., 2024; Sang et al., 2025b). Attribute extraction, that is, the original data is projected into a low-dimensional space to remove irrelevant and redundant attributes from the data. However, it makes the low-dimensional data no longer have the same physical meaning as before, which will probably affect its application in real life. Attribute selection removes irrelevant and redundant attributes from the original data by selecting an optimal subset of attributes without destroying the data's physical meaning. In this paper, we focus on the study of attribute selection.

Attribute selection methods can be broadly classified into three categories based on the presence or absence of labeling information in the data (Chen et al., 2022b; Liao et al., 2024; Sang et al., 2025a; Yuan et al., 2021a), which are unsupervised, semi-supervised, and supervised methods. For semi-supervised or supervised methods, we must know part or all of the sample's labeling information in advance. In practical contexts, obtaining the sample's labeling information is exceptionally expensive. Unsupervised attribute selection (UAS) can identify an optimal subset of attributes based on the intrinsic structure of the data. Therefore, it is essential to study UAS. Most of the existing UAS methods are based on studying spectral analysis and manifold learning for attribute selection,

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which can preserve the intrinsic structure of the data. However, to the best of our knowledge, most of the existing methods are currently unable to reflect the granular knowledge structure, uncertainty, fuzziness, and inconsistency in the data. And they only apply to a single attribute type (nominal or numerical) (Yuan et al., 2021c). But, in practical applications, the data is usually in uncertain, hybrid (nominal and numerical), and unlabeled forms. For instance, the medical diagnostic data usually contains: name, sex, race, weight, height, blood pressure, etc. Where weight, height, and blood pressure are numerical attributes; name, sex, and race are nominal attributes. Nowadays, the volume of uncertain hybrid unlabeled data is enormous. Therefore, it is necessary to reduce the data dimensionally to decrease the computational overhead. UAS is an effective method to reduce the data dimensionally and is worth studying in uncertain hybrid data.

The rough set theory (RST) is a mathematical framework for handling uncertain hybrid data (Pawlak & Skowron, 2007), which has been successfully applied in many fields, for instance, attribute selection (Sang et al., 2025a,b; Yang et al., 2024; Zhang et al., 2024), outlier analysis (Chen et al., 2024; Hu et al., 2025; Su et al., 2024; Zhang et al., 2023a), classification task (Li et al., 2023a; Yue et al., 2020), rule induction (Zhao et al., 2021), and other aspects (Wang et al., 2022). Attribute selection is one of the main research aspects of RST. It aims to identify an optimal subset of attributes from the high-dimensional attribute space without decreasing the classification ability of the data (Zhang et al., 2022b, 2019; Zhu et al., 2018, 2015). But, it applies only to handling nominal attribute data based on equivalence relations. When it handles numerical attribute data, it needs to be discretized. However, the discretization process will lead to the loss of data information (Wang et al., 2023; Zhang et al., 2022a). In response to the above problems, Hu et al. proposed the neighborhood rough set (NRS) model (Hu et al., 2008a,b,c). NRS provides an effective tool to overcome the discretization problem. And it can be directly applied to attribute selection of numerical attribute data without discretization. In NRS, the neighborhood relations are used to measure the similarity between the data samples. Currently, research on NRS focuses mainly on the extension and application of the NRS model. On the one hand, inspired by the original NRS model, a series of NRS extension models have been proposed (Jiang et al., 2024; Liu et al., 2022; Xia et al., 2022; Xie et al., 2022); On the other hand, NRS has been successfully applied to attribute selection (Liu et al., 2024; Wang & Zhao, 2024; Wu et al., 2024; Zhang et al., 2022a), multi-label attribute selection (Lin et al., 2016; Liu et al., 2018; Sun et al., 2020), hyperspectral analysis (Liu et al., 2016; Yu et al., 2019), anomaly detection (Wang & Li, 2021; Yuan et al., 2018), and other fields. It presents better research value and application prospects.

Attribute selection is also called attribute reduction in the context of rough sets (Yuan et al., 2021a). Recently, attribute selection methods based on neighborhood rough sets (NRSs) have received much attention. In general, attribute selection methods based on NRSs can be classified into three main categories. They are neighborhood dependency-based (Li et al., 2024b), neighborhood discriminative matrix-based (Wang et al., 2018), and neighborhood uncertainty measures-based (Zhang et al., 2022a) methods. Li et al. proposed UAS methods based on neighborhood dependency (Li et al., 2024b) and variable precision weighted neighborhood dependency (Li et al., 2024a), respectively. However, these methods apply only to numerical attribute data. Cheng et al. proposed a hybrid ordered data attribute selection method based on dominant NRS (Chen & Zhu, 2024), which utilized a neighborhood discriminative matrix. Zhang et al. proposed the attribute selection method based on neighborhood combination entropy (Zhang et al., 2022a). To handle uncertain hybrid attribute data effectively, Hu et al. proposed attribute selection methods based on neighborhood dependency (Hu et al., 2008a) and neighborhood information entropy (Hu & Yu, 2009), respectively. They achieved good results.

To the best of our knowledge, the majority of the NRS attribute selection methods proposed above are supervised and cannot be applied to unsupervised tasks. Based on this, we propose an unsupervised hybrid attribute selection model based on neighborhood attribute quality degree. Its computational model is shown in Fig. 1. First, we construct the variable precision neighborhood rough set (VPNRS) model for the decision-free information system. Then, we compute the variable precision neighborhood dependency (VPND) and the increase rate of the average variable precision neighborhood correct classification rate (VPNCCR) based on all single-condition attribute subsets to define the attribute quality degree. Based on this, we define the neighborhood attribute quality degree, which can be used as an important metric for evaluating candidate attributes. Next, an unsupervised hybrid attribute selection algorithm based on neighborhood attribute quality degree (UAS_NAQD) is designed. Finally, UAS_NAQD is compared and analyzed with existing UAS algorithms using public datasets. Experimental results show that UAS_NAQD can choose fewer attributes to maintain or improve the performance of clustering algorithms. In addition, the statistical test shows that UAS_NAQD proposed in this paper is statistically significantly different from most existing UAS algorithms. In summary, the main contributions of this paper are as follows.

(1) Based on NRS theory, the idea of variable precision is introduced to construct the VPNRS model for the decision-free information system. This model has better performance of noise resistance and fault tolerance than NRS model.
(2) To effectively handle uncertain hybrid attribute data, we have defined the neighborhood attribute quality degree based on the VPND and the increase rate of the average VPNCCR w.r.t. all single-condition attribute subsets. Based on this, we have designed UAS_NAQD.
(3) We have performed several experiments on 32 different types of high-dimensional datasets. Experimental results show that UAS_NAQD proposed in this paper can identify an optimal subset of attributes to maintain or improve the performance of clustering algorithms. In addition, the statistical test shows that UAS_NAQD is statistically significantly different from 9 other existing UAS algorithms.

The remainder of this paper is organized as follows. Section 2 presents related work. Section 3 presents the basic theory of variable precision neighborhood rough sets (VPNRSs). Section 4 gives an unsupervised hybrid attribute selection model based on neighborhood attribute quality degree and also specifically analyzes the model. The corresponding algorithm UAS_NAQD is proposed. Section 5 presents the experimental results and analysis. Section 6 concludes the paper.

2. Related work

In this section, we will briefly present some of the latest studies on UAS and methods of attribute selection based on NRS.

UAS can be broadly classified into three categories (Chen et al., 2022b; Liao et al., 2024). They are filter, wrapper, and embedded methods. The filter methods evaluate each attribute's significance individually, ignoring the correlation between attributes (Li et al., 2023b). Although they are computationally efficient, they may not perform well. It consists of three main methods (Solorio-Fernández et al., 2020). They are bio-inspired, statistical information, and spectral analysis methods. Tabakhi proposed a UAS method based on ant colony optimization, called UFSACO (Tabakhi et al., 2014). The wrapper methods combine attribute selection with a specific learning task. They aim to identify an optimal subset of attributes that perform optimally in the task's execution. Although their performance is better, they have high computational overhead and the risk of falling into overfitting (Li et al., 2021). It consists of three main methods (Solorio-Fernández et al., 2020). They are sequential, bio-inspired, and iterative methods. Dutta et al. proposed the UAS method based on bio-inspired (Dutta et al., 2014). The embedded methods integrate attribute selection and model training into a unified framework. They identify the attribute that contributes the most during the training process. Embedded methods are often efficient and

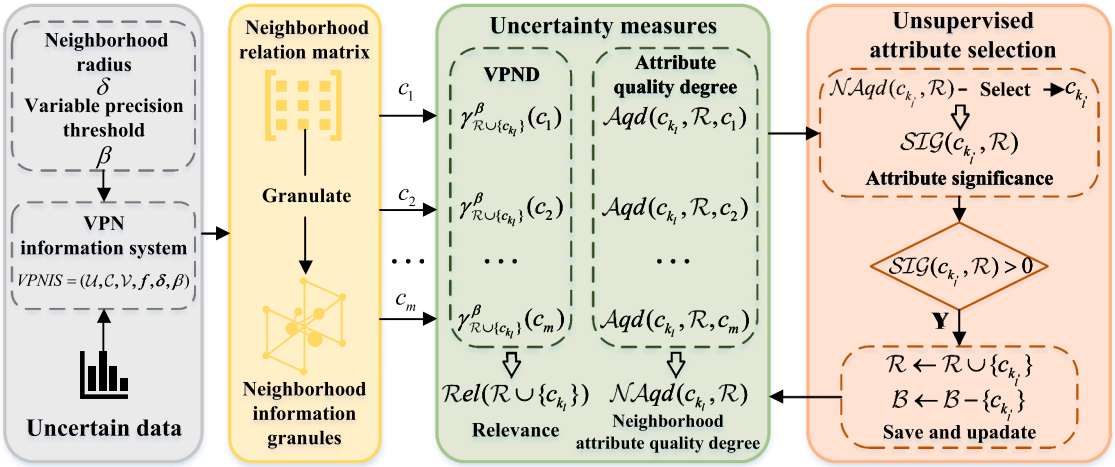


Fig. 1. The computational model of UAS_NAQD.

effective. Therefore, many scholars have been exploring UAS methods based on embedded. They have done a lot of work. For instance, Liao et al. proposed a general adaptive UAS method with auto-weighting (Liao et al., 2024); Chen et al. proposed a fast UAS method based on the bipartite graph and $\ell_{2,1}$-norm constraint (Chen et al., 2022a); Chen et al. proposed the UAS method based on the flexible optimal graph and $\ell_{2,1}$-norm regularization (Chen et al., 2022b). In addition, there are some other prominent UAS methods. For instance, Li et al. proposed a new UAS method (Li et al., 2021), which aims to improve the accuracy and efficiency of attribute selection by simultaneously learning graph structure, attribute weights, and projection matrix, as well as introducing the concept of adaptive neighbors. Yuan et al. proposed the UAS method based on the fuzzy complementary entropy of hybrid-kernel function (Yuan et al., 2021b), which is devoted to handling uncertain hybrid data. Through literature, we found that most UAS methods apply only to a single type of data (nominal or numerical) (Li et al., 2024b; Zhang et al., 2023b). There have been only a few attempts at UAS problems for uncertain hybrid data (Yuan et al., 2021a; Zhang et al., 2024). For instance, Yuan et al. proposed a UAS method based on fuzzy dependency (Yuan et al., 2021a); Zhang et al. proposed a UAS method, which integrated multi-scale information and considered multiple correlations (Zhang et al., 2024); Yang et al. proposed a UAS method based on fuzzy multi-neighborhood entropy (Yang et al., 2024).

However, most of the above methods can not handle uncertain hybrid data. NRS can effectively handle uncertain hybrid data. The classical rough set model describes the imprecise concept by the upper and lower approximation sets based on the equivalence relation. Hu et al. introduced the neighborhood relation into the classical rough set model and proposed the NRS model (Hu et al., 2008a), which can handle hybrid attribute data. In recent years, many scholars have carried out further studies in the framework of NRSs. For lower approximation sets of NRS, which only allow samples to be strictly contained. This harsh partition condition makes NRS less tolerant of noise. At the same time, it is too sensitive to miss-classification cases. Hu et al. introduced the variable precision proposed by Ziarko (Ziarko, 1993) into NRS and proposed the VPNRS (Hu et al., 2008a) model. Based on this, Li et al. proposed a UAS method based on variable precision weighted neighborhood dependency (Li et al., 2024a). Although this method considers the weight of samples, it applies only to numerical attribute data. Zhang et al. proposed a streaming attribute selection method based on NRSs (Zhang et al., 2025), which considers the key role of attribute interaction in classification accuracy. Wang et al. proposed a new attribute selection method based on weighted k-nearest NRS (Wang & Zhao, 2024), which solves the problem of noise sensitivity of NRSs. Sun et al. proposed an online group streaming attribute selection method based on fuzzy neighborhood granular ball rough sets (Sun & Zhu, 2024), which not only achieves parameter-free design but also improves classification efficiency. Xia et al. proposed an attribute selection method based on Granular-Ball NRS (Xia et al., 2025). Wan et al. proposed a novel hybrid attribute selection method considering attribute interaction in NRS (Wan et al., 2021). At the same time, many scholars have integrated intelligent algorithms into NRS attribute selection methods and have achieved significant results. Liu et al. proposed an adaptive NRS based on the sparrow search algorithm and applied the model to attribute selection (Liu et al., 2024). Sun et al. proposed a two-stage attribute selection method based on via fusion fuzzy multi-NRS with binary whale optimization (Sun et al., 2023).

In summary, scholars have done a lot of work based on NRS attribute selection. However, it is difficult for these methods to handle uncertain hybrid data. Therefore, it is of great theoretical value and practical significance to study UAS for uncertain hybrid data based on NRSs.

3. Preliminaries

In this section, some knowledge about VPNRSs will be introduced (Hu et al., 2008a).

Definition 1. Hu et al. (2008b,c) A five-tuple can be used to represent neighborhood information system (NIS), i.e., $NIS = (\mathcal{U}, \mathcal{A}, \mathcal{V}, f, \delta)$, where $\mathcal{U} = \{x_1, x_2, \cdots, x_n\}$ is a non-empty sample set; $\mathcal{A} = \{a_1, a_2, \cdots, a_m\}$ is a non-empty attribute set; $\mathcal{V} = \cup \mathcal{V}_a (\forall a \in \mathcal{A})$ is the union of attribute values of the samples in $\mathcal{U}$. $\mathcal{V}_a$ is the domain of values of the samples in $\mathcal{U}$ under attribute a; $f : \mathcal{U} \times \mathcal{A} \to \mathcal{V}$ is an information function, for $\forall x \in \mathcal{U}$ and $\forall a \in \mathcal{A}$, which has $f(x, a) \in \mathcal{V}_a$. It represents the information relation between the sample and the attribute to the attribute value. $\delta (\delta \geq 0)$ is the neighborhood radius parameter for neighborhood granulation of the samples in $\mathcal{U}$.

In NIS, $\mathcal{A}$ is composed of the conditional attribute set $\mathcal{C}$ and the decision attribute set $\mathcal{D}$, i.e., $\mathcal{A} = \mathcal{C} \cup \mathcal{D} (\mathcal{C} \cap \mathcal{D} = \emptyset)$. If $\mathcal{D} \neq \emptyset$, at this point, NIS is a decision neighborhood information system (DNIS), i.e., $DNIS = (\mathcal{U}, \mathcal{C} \cup \mathcal{D}, \mathcal{V}, f, \delta)$.

Given a $NIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta)$, the distance function $\triangle$ and neighborhood radius parameter δ are introduced to measure the discriminability of the samples under the conditional attributes. For $\forall \mathcal{B} \subseteq \mathcal{C}$, $\mathcal{C} = \{c_1, c_2, \cdots, c_m\}$, and $\mathcal{B} = \{c_{k_1}, c_{k_2}, \cdots, c_{k_h}\} (1 \leq h \leq m)$. Let $\forall x_i, x_j, x_k \in \mathcal{U} (1 \leq i, j, k \leq n)$, the distance function $\triangle_{\mathcal{B}}(x_i, x_j)$ satisfies:

(1) Nonnegative: $\triangle_{\mathcal{B}}(x_i, x_j) \geq 0, \triangle_{\mathcal{B}}(x_i, x_i) = 0$;
(2) Symmetry: $\triangle_{\mathcal{B}}(x_i, x_j) = \triangle_{\mathcal{B}}(x_j, x_i)$;
(3) Triangle Inequality: $\triangle_{\mathcal{B}}(x_i, x_k) \leq \triangle_{\mathcal{B}}(x_i, x_j) + \triangle_{\mathcal{B}}(x_j, x_k)$.

$\triangle_{\mathcal{B}}(x_i, x_j)$ denotes the distance between x_i and x_j under $\mathcal{B}$. Next, the δ-neighborhood will be defined.

Definition 2. Hu et al. (2008b,c) Given a $NIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta)$, for $\forall x_i \in \mathcal{U}$ and $\forall \mathcal{B} \subseteq \mathcal{C}$, the δ-neighborhood of x_i under $\mathcal{B}$ is defined as

$$n_{\mathcal{B}}^{\delta}(x_i) = \{x_j \in \mathcal{U} | \triangle_{\mathcal{B}}(x_i, x_j) \le \delta\}, \tag{1}$$

where $\delta \ge 0$, $n_{\mathcal{B}}^{\delta}(x_i)$ is called the δ-neighborhood by x_i under $\mathcal{B}$. It is also called the neighborhood information granule of x_i under $\mathcal{B}$. $n_{\mathcal{B}}^{\delta}(x_i)$ denotes the sample set in $\mathcal{U}$ under $\mathcal{B}$ of x_i. The samples in the sample set are indistinguishable under $\mathcal{B}$.

When nominal and numerical attribute data coexist in NIS, the δ-neighborhood of x_i needs to be redefined. Given a $NIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta)$, for $\forall \mathcal{B}_1, \mathcal{B}_2 \in \mathcal{B}$. If $\mathcal{B}_1 \cup \mathcal{B}_2 = \mathcal{B}$ and $\mathcal{B}_1 \cap \mathcal{B}_2 = \emptyset$, the δ-neighborhood of x_i can be redefined as

(1) $n_{\mathcal{B}_1}^{\delta}(x_i) = \{x_j \in \mathcal{U} | \triangle_{\mathcal{B}_1}(x_i, x_j) = 0\}$;
(2) $n_{\mathcal{B}_2}^{\delta}(x_i) = \{x_j \in \mathcal{U} | \triangle_{\mathcal{B}_2}(x_i, x_j) \le \delta\}$;
(3) $n_{\mathcal{B}_1 \cup \mathcal{B}_2}^{\delta}(x_i) = \{x_j \in \mathcal{U} | \triangle_{\mathcal{B}_1}(x_i, x_j) = 0, \triangle_{\mathcal{B}_2}(x_i, x_j) \le \delta\}$.

It can be seen that when the neighborhood information granule of x_i is constructed, for the nominal attribute set, we generally take $\delta = 0$; for the numerical attribute set, we generally take $\delta > 0$.

Property 1. *Hu et al. (2008b,c) Given a $NIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta)$, for $\forall x_i \in \mathcal{U}$ and $\forall \mathcal{P}, \mathcal{Q} \subseteq \mathcal{C}$, the δ-neighborhood of x_i has the following properties.*

(1) $\forall \mathcal{P} \subseteq \mathcal{Q} \Rightarrow n_{\mathcal{P}}^{\delta}(x_i) \supseteq n_{\mathcal{Q}}^{\delta}(x_i)$;
(2) $\forall 0 \le \delta_1 \le \delta_2 \Rightarrow n_{\mathcal{P}}^{\delta_1}(x_i) \subseteq n_{\mathcal{P}}^{\delta_2}(x_i)$.

Property 1(1) shows that the inclusion relation of $n_{\mathcal{P}}^{\delta}(x_i)$ and $n_{\mathcal{Q}}^{\delta}(x_i)$ is opposite to the inclusion relation of $\mathcal{P}$ and $\mathcal{Q}$. Property 1(2) shows that the size relation of δ_1 and δ_2 remains the same as the inclusion relation of $n_{\mathcal{P}}^{\delta_1}(x_i)$ and $n_{\mathcal{P}}^{\delta_2}(x_i)$.

Property 2. *Hu et al. (2008b,c) Given a $NIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta)$, for $\forall x_i, x_j \in \mathcal{U}$ and $\forall \mathcal{B} \subseteq \mathcal{C}$. $n_{\mathcal{B}}^{\delta}(x_i)$ has following properties:*

(1) $x_i \in n_{\mathcal{B}}^{\delta}(x_i) \Rightarrow n_{\mathcal{B}}^{\delta}(x_i) \ne \emptyset$;
(2) $x_j \in n_{\mathcal{B}}^{\delta}(x_i) \Leftrightarrow x_i \in n_{\mathcal{B}}^{\delta}(x_j)$;
(3) $1 \le |n_{\mathcal{B}}^{\delta}(x_i)| \le |\mathcal{U}|$;
(4) $\bigcup_{x_i \in \mathcal{U}} n_{\mathcal{B}}^{\delta}(x_i) = \mathcal{U}$.

Based on δ-neighborhood, we will give the definition of the neighborhood relation $\mathcal{NR}_{\mathcal{B}}$ of the samples in $\mathcal{U}$ under $\mathcal{B}$.

Definition 3. Hu et al. (2008b,c) Given a $NIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta)$, for $\forall x_i, x_j \in \mathcal{U}$ and $\forall \mathcal{B} \subseteq \mathcal{C}$, the neighborhood relation is defined as

$$\mathcal{NR}_{\mathcal{B}} = \{(x_i, x_j) \in \mathcal{U} \times \mathcal{U} | \triangle_{\mathcal{B}}(x_i, x_j) \le \delta\}. \tag{2}$$

$\mathcal{NR}_{\mathcal{B}}$ is a similarity relation based on the distance measure $\triangle$, which extends the applicable scope of the equivalence relation. When $\delta = 0$, $\mathcal{B}$ is the nominal attribute set, $\mathcal{NR}_{\mathcal{B}}$ degenerates to an equivalence relation; when $\delta > 0$, $\mathcal{B}$ is the numerical attribute set, which can handle numerical attribute data.

Property 3. *Hu et al. (2008b,c) Given a $NIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta)$, for $\forall \mathcal{P}, \mathcal{Q} \subseteq \mathcal{C}$, the property of the neighborhood relation is $\forall \mathcal{P} \subseteq \mathcal{Q} \Rightarrow \mathcal{NR}_{\mathcal{P}} \supseteq \mathcal{NR}_{\mathcal{Q}}$.*

By NIS granulation, the neighborhood information granule can be obtained.

Definition 4. Hu et al. (2008b,c) Given a $NIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta)$, for $\forall \mathcal{B} \subseteq \mathcal{C}$, $\mathcal{NR}_{\mathcal{B}}$ is the neighborhood relation on $\mathcal{U}$. For $\forall \mathcal{X} \subseteq \mathcal{U}$, the upper approximation set $\overline{\mathcal{NR}_{\mathcal{B}}}(\mathcal{X})$ and the lower approximation set $\underline{\mathcal{NR}_{\mathcal{B}}}(\mathcal{X})$ of $\mathcal{X}$ are defined respectively as

$$\overline{\mathcal{NR}_{\mathcal{B}}}(\mathcal{X}) = \{x \in \mathcal{U} | n_{\mathcal{B}}^{\delta}(x) \cap \mathcal{X} \ne \emptyset\}, \tag{3}$$

$$\underline{\mathcal{NR}_{\mathcal{B}}}(X) = \{x \in \mathcal{U} | n_{\mathcal{B}}^{\delta}(x) \subseteq \mathcal{X}\}. \tag{4}$$

Where, we call $\mathcal{POS}_{\mathcal{NR}_{\mathcal{B}}}(\mathcal{X}) = \underline{\mathcal{NR}_{\mathcal{B}}}(\mathcal{X})$ the positive region of $\mathcal{X}$; $\mathcal{NEG}_{\mathcal{NR}_{\mathcal{B}}}(\mathcal{X}) = \mathcal{U} - \overline{\mathcal{NR}_{\mathcal{B}}}(\mathcal{X})$ the negative region of $\mathcal{X}$; $\mathcal{BND}_{\mathcal{NR}_{\mathcal{B}}}(\mathcal{X}) = \overline{\mathcal{NR}_{\mathcal{B}}}(\mathcal{X}) - \underline{\mathcal{NR}_{\mathcal{B}}}(\mathcal{X})$ the boundary region of $\mathcal{X}$.

Next, based on the relevant knowledge of NRS, the VPNRS will be defined.

Definition 5. (Hu et al., 2008a) Given a $NIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta)$, the variable precision threshold parameter $\beta (0.5 \le \beta \le 1)$ is introduced. Then, the VPNIS model can be represented by a six-tuple, i.e., $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$. $(\mathcal{U}, \mathcal{C} \cup \mathcal{D}, \mathcal{V}, f, \delta, \beta)$ is called the variable precision decision neighborhood information system (VPDNIS).

Based on Definition 5, we will give definitions of the upper and lower approximation sets of the variable precision neighborhood (VPN) of the sample in $\mathcal{U}$.

Definition 6. Hu et al. (2008a) Given a $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, for $\forall \mathcal{X} \subseteq \mathcal{U}$, the upper approximation set of the VPN $\overline{\mathcal{NR}_{\mathcal{B}}^{\beta}}(\mathcal{X})$ and the lower approximation set of the VPN $\underline{\mathcal{NR}_{\mathcal{B}}^{\beta}}(\mathcal{X})$ based on the neighborhood relation $\mathcal{NR}_{\mathcal{B}}$ are defined respectively as

$$\overline{\mathcal{NR}_{\mathcal{B}}^{\beta}}(\mathcal{X}) = \{x \in \mathcal{U} | \frac{|n_{\mathcal{B}}^{\delta}(x) \cap \mathcal{X}|}{|n_{\mathcal{B}}^{\delta}(x)|} \ge 1 - \beta\}, \tag{5}$$

$$\underline{\mathcal{NR}_{\mathcal{B}}^{\beta}}(\mathcal{X}) = \{x \in \mathcal{U} | \frac{|n_{\mathcal{B}}^{\delta}(x) \cap \mathcal{X}|}{|n_{\mathcal{B}}^{\delta}(x)|} \ge \beta\}. \tag{6}$$

In NRS, by introducing the variable precision threshold parameter β, which relaxes the strict definition of the positive region. The size of β is inversely related to the size of the positive region, i.e., β is smaller, the positive region is larger. Therefore, VPNRS has a certain tolerance to uncertain hybrid data and has a good performance in noise resistance. In addition, when $\beta = 1$, VPNIS will degenerate to standard NIS.

4. Proposed method

In this section, a new VPNRS considering conditional attributes is first built. Next, an unsupervised hybrid attribute selection model based on neighborhood attribute degree is proposed. Then, the corresponding attribute selection algorithm UAS_NAQD is designed. Finally, a specific example is given for illustration.

4.1. VPNRS model considering conditional attributes

Given a $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, the nominal attribute set $\mathcal{B}_1 \subseteq \mathcal{B}$, the numerical attribute set $\mathcal{B}_2 \subseteq \mathcal{B}$, $\mathcal{B}_1 \cup \mathcal{B}_2 = \mathcal{B}$, and $\mathcal{B}_1 \cap \mathcal{B}_2 = \emptyset$. To efficiently handle nominal, numerical, and hybrid attribute data. For $\forall x_i, x_j \in \mathcal{U}$, $\mathcal{NR}_{\mathcal{B}}$ under $\mathcal{B}$ can be represented by $r_{ij}^{\mathcal{B}}$, which is defined as

$$r_{ij}^{\mathcal{B}} = \begin{cases} 1, \text{if } \triangle_{\mathcal{B}}(x_i, x_j) = 0, \mathcal{B} = \mathcal{B}_1, \text{ and } \mathcal{B}_2 = \emptyset; \\ 0, \text{if } \triangle_{\mathcal{B}}(x_i, x_j) > 0, \mathcal{B} = \mathcal{B}_1, \text{ and } \mathcal{B}_2 = \emptyset; \\ 1, \text{if } \triangle_{\mathcal{B}}(x_i, x_j) \le \delta_{\mathcal{B}}, \mathcal{B} = \mathcal{B}_2, \text{ and } \mathcal{B}_1 = \emptyset; \\ 0, \text{if } \triangle_{\mathcal{B}}(x_i, x_j) > \delta_{\mathcal{B}}, \mathcal{B} = \mathcal{B}_2, \text{ and } \mathcal{B}_1 = \emptyset. \end{cases} \tag{7}$$

In this paper, for the nominal attribute set $\mathcal{B}_1$, we use Hamming Distance (Yuan et al., 2018); for the numerical attribute set $\mathcal{B}_2$, we use the Euclidean Distance (Hu et al., 2008a). From Eq. (7), it can be seen that when the distance between x_i and x_j is less than δ or equal to 0, we consider them indistinguishable; when the distance between x_i and x_j is greater than δ, we consider them to be distinguishable. This approach reflects people's tolerance for the noise of hybrid uncertain data, where δ is adaptive. Next, the $\delta_{\mathcal{B}}$ will be defined.

Definition 7. Given a $VPNIS = (\mathcal{U}, \mathcal{V}, \mathcal{C}, f, \delta, \beta)$, for $\forall \mathcal{B} \subseteq \mathcal{C}$, let $\mathcal{B} = \{c_{k_1}, c_{k_2}, \cdots, c_{k_h}\} (0 \le h \le m)$. The nominal attribute set $\mathcal{B}_1 = \{c_{i_1}, c_{i_1}, \cdots, c_{i_l}\} (0 \le l \le h)$, the numerical attribute set

$\mathcal{B}_2 = \{c_{j_1}, c_{j_2}, \cdots, c_{j_q}\}(0 \le q \le h)$, $\mathcal{B}_1 \cup \mathcal{B}_2 = \mathcal{B}$, and $\mathcal{B}_1 \cap \mathcal{B}_2 = \emptyset$, the formula for computing $\delta_{\mathcal{B}_2}$ is defined as

$$\delta_{\mathcal{B}_2} = \frac{\sum_{k=1}^{q} std(c_{j_k})}{|\mathcal{B}_2|}, \tag{8}$$

where $std(j_k)$ is the standard deviation of the value of c_{j_k}. $|\mathcal{B}_2|$ is the number of elements in $\mathcal{B}_2$.

$\mathcal{NR}_{\mathcal{B}}$ is a neighborhood relation of the samples in $\mathcal{U}$ w.r.t. $\mathcal{B}$, the neighborhood relation matrix $\mathcal{M}(\mathcal{NR}_{\mathcal{B}})$ is defined as

$$\mathcal{M}(\mathcal{NR}_{\mathcal{B}}) = \begin{pmatrix} r^{\mathcal{B}}_{11} & r^{\mathcal{B}}_{12} & \cdots & r^{\mathcal{B}}_{1n} \\ r^{\mathcal{B}}_{21} & r^{\mathcal{B}}_{22} & \cdots & r^{\mathcal{B}}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r^{\mathcal{B}}_{n1} & r^{\mathcal{B}}_{n2} & \cdots & r^{\mathcal{B}}_{nn} \end{pmatrix}. \tag{9}$$

It follows that the δ-neighborhood of x_i in $\mathcal{U}$ can be represented as a 0–1 vector form under $\mathcal{B}$, i.e., $n^{\delta}_{\mathcal{B}}(x_i) = (r^{\mathcal{B}}_{i1}, r^{\mathcal{B}}_{i2}, \cdots, r^{\mathcal{B}}_{in})$. Based on this, the family of neighborhood information granules will be defined.

Definition 8. The family of neighborhood information granules of the samples in $\mathcal{U}$ is generated by $\mathcal{NR}_{\mathcal{B}}$, which is defined as

$$\mathcal{G}(\mathcal{NR}_{\mathcal{B}}) = \{n^{\delta}_{\mathcal{B}}(x_1), n^{\delta}_{\mathcal{B}}(x_2), \cdots, n^{\delta}_{\mathcal{B}}(x_n)\}. \tag{10}$$

$n^{\delta}_{\mathcal{B}}(x_i)$ is the neighborhood information granule induced by $\mathcal{NR}_{\mathcal{B}}$ and contains x_i. Without causing confusion, we use $\mathcal{B}$ instead of $\mathcal{NR}_{\mathcal{B}}$, i.e., $\mathcal{G}(\mathcal{NR}_{\mathcal{B}})$ can be written simply as $\mathcal{G}(\mathcal{B})$.

In VPNIS, the upper and lower approximation sets of the VPN of neighborhood information granules are proposed for decision attribute. Based on this, this paper will give the definitions of the upper and lower approximation sets of the VPN for conditional attributes.

Definition 9. For $\forall \mathcal{P}, \mathcal{Q} \subseteq \mathcal{C}$, let $\mathcal{G}(\mathcal{P}) = \{n^{\delta}_{\mathcal{P}}(x_1), n^{\delta}_{\mathcal{P}}(x_2), \cdots, n^{\delta}_{\mathcal{P}}(x_n)\}$ and $\mathcal{G}(\mathcal{Q}) = \{n^{\delta}_{\mathcal{Q}}(x_1), n^{\delta}_{\mathcal{Q}}(x_2), \cdots, n^{\delta}_{\mathcal{Q}}(x_n)\}$. For $\forall n^{\delta}_{\mathcal{P}}(x_i) \in \mathcal{G}(\mathcal{P})$, $\forall n^{\delta}_{\mathcal{Q}}(x_j) \in \mathcal{G}(\mathcal{Q})$, and β, the upper and lower approximation sets of the VPN of $n^{\delta}_{\mathcal{Q}}(x_j)$ w.r.t. $\mathcal{NR}_{\mathcal{P}}$ are defined respectively as

$$\overline{\mathcal{NR}^{\beta}_{\mathcal{P}}}(n^{\delta}_{\mathcal{Q}}(x_j)) = \{x_i \in \mathcal{U} | \frac{|n^{\delta}_{\mathcal{P}}(x_i) \cap n^{\delta}_{\mathcal{Q}}(x_j)|}{|n^{\delta}_{\mathcal{P}}(x_i)|} \ge 1 - \beta\}, \tag{11}$$

$$\underline{\mathcal{NR}^{\beta}_{\mathcal{P}}}(n^{\delta}_{\mathcal{Q}}(x_j)) = \{x_i \in \mathcal{U} | \frac{|n^{\delta}_{\mathcal{P}}(x_i) \cap n^{\delta}_{\mathcal{Q}}(x_j)|}{|n^{\delta}_{\mathcal{P}}(x_i)|} \ge \beta\}. \tag{12}$$

The upper and lower approximation sets of the VPN are based on the fault tolerance partition of β, which allows a certain amount of misclassification. The size of β determines the sample coverage of the upper and lower approximation sets.

Dependency is an important concept in classical RST, which utilizes the positive region to analyze the degree of dependence between conditional attributes. Currently, it is widely used in supervised attribute selection. Based on this, to describe the relevance between conditional attributes, we will define the dependency in VPNRS to measure the relevance of conditional attributes. To this end, we first give the concept of the positive region of the VPN.

Definition 10. For $\forall \mathcal{Q}, \mathcal{P} \subseteq \mathcal{C}$, the positive region of the VPN of $\mathcal{Q}$ w.r.t. $\mathcal{P}$ is defined as

$$\mathcal{VPNPOS}^{\beta}_{\mathcal{P}}(\mathcal{Q}) = \bigcup_{n^{\delta}_{\mathcal{Q}}(x_j) \in \mathcal{G}(\mathcal{Q})} \underline{\mathcal{NR}^{\beta}_{\mathcal{P}}}(n^{\delta}_{\mathcal{Q}}(x_j)). \tag{13}$$

The positive region of the VPN describes the union of the lower approximation set of the VPN of $\mathcal{Q}$ w.r.t. $\mathcal{P}$. Its size reflects the relevance of conditional attributes.

Based on Definition 10, the VPND will be defined. In most cases, conditional attributes are partially relevant to each other. This means that $\mathcal{Q}$ is only relevant to $\mathcal{P}$ to a certain extent. The VPND can describe partial relevance.

Definition 11. For $\forall \mathcal{Q}, \mathcal{P} \subseteq \mathcal{C}$, the VPND of $\mathcal{Q}$ w.r.t. $\mathcal{P}$ is defined as

$$\gamma^{\beta}_{\mathcal{P}}(\mathcal{Q}) = \frac{|\mathcal{VPNPOS}^{\beta}_{\mathcal{P}}(\mathcal{Q})|}{|\mathcal{U}|}. \tag{14}$$

$0 \le \gamma^{\beta}_{\mathcal{P}}(\mathcal{Q}) \le 1$, which reflects the relevance of the conditional attributes to each other. When $\gamma^{\beta}_{\mathcal{P}}(\mathcal{Q}) = 1$, we call $\mathcal{Q}$ completely dependent on $\mathcal{P}$; When $0 < \gamma^{\beta}_{\mathcal{P}}(\mathcal{Q}) < 1$, we call $\mathcal{Q}$ partially dependent on $\mathcal{P}$; When $\gamma^{\beta}_{\mathcal{P}}(\mathcal{Q}) = 0$, we call $\mathcal{Q}$ completely independent on $\mathcal{P}$. We want to find a subset of attributes with the largest dependency. In this case, the attributes in the subset of attributes are more relevant to each other. Based on Definition 11, the attribute's significance will be defined.

Definition 12. Given a $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, for $\forall \mathcal{P}, \mathcal{Q} \subseteq \mathcal{C}$ and $\forall c_k \in \mathcal{C} - \mathcal{P}(1 \le k \le m)$, the significance of the attribute c_k w.r.t. $\mathcal{P}$ considering $\mathcal{Q}$ is defined as

$$Sig(c_k, \mathcal{P}, \mathcal{Q}) = \gamma^{\beta}_{\mathcal{P} \cup \{c_k\}}(\mathcal{Q}) - \gamma^{\beta}_{\mathcal{P}}(\mathcal{Q}). \tag{15}$$

Based on the analysis of Definitions 11 and 12, we know that the attribute's significance depends on the change in dependency, which depends on the partition of the positive region. If $Sig(c_k, \mathcal{P}, \mathcal{Q}) > 0$, which shows that when the attribute c_k is added, the coverage of the partition of the positive region by the new subset of attributes $\mathcal{P} \cup \{c_k\}$ will increase. We can rely on this new subset of attributes $\mathcal{P} \cup \{c_k\}$ for more accurate classification.

In previous attribute selection studies, most of the methods are forward search algorithms, which use the attribute's significance as the heuristic factor (Li et al., 2024b; Yuan et al., 2021a). Attribute selection methods based on VPNRS also adopt this idea (Hu et al., 2008a; Li et al., 2024a). However, these methods have certain disadvantages due to the introduced β.

From Definition 9, VPNRS can adjust β to control the partition of the positive region, which can influence the result of attribute selection. The smaller the value of β, the more relaxed the partition condition of the positive region, the coverage of the positive region increases, and the number of selected attributes will increase; The larger the value of β, the more strict the partition condition of the positive region, the coverage of the positive region decreases, and the number of selected attributes will decrease. For different datasets, to obtain the optimal result, the value of β is uncertain. During the experiment, we need to keep adjusting the value of β to find the most suitable value. However, in the process of calculating the attribute's significance, they ignored the effect of β on the positive region. To address this problem, we need to optimize the attribute's significance of Eq. (15).

In general, Hu et al. neglected the effect of β on the positive region when they computed the attribute's significance (Hu et al., 2008a). This makes the credibility of the attribute's significance less, which may cause the attribute with poor classification ability to be selected into the reduced set. So, how to introduce β to improve fault tolerance while avoiding the effect of β on the positive region. This is one of the main problems to be solved in this paper. We will optimize the attribute's significance, which was proposed by Hu et al. (Hu et al., 2008a). First, we will define the increase rate of the average VPNCCR of the sample within the positive region of the VPND based on conditional attributes. Then, it is used as the quality factor of the attribute's significance to define the attribute quality degree. Next, based on attribute quality degree to define the neighborhood attribute quality degree. Finally, an unsupervised hybrid attribute selection algorithm based on neighborhood attribute quality degree is proposed, which can preferentially select the attribute with better classification ability.

Based on this, we will first give the definition of the VPNCCR of the samples within the positive region of the VPN based on conditional attributes.

Definition 13. Given a $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, for $\forall \mathcal{P}, \mathcal{Q} \in \mathcal{C}$, $\forall x_i \in \mathcal{VPNPOS}^{\beta}_{\mathcal{P}}(\mathcal{Q})$ and $n^{\delta}_{\mathcal{P}}(x_i) \in \mathcal{G}(\mathcal{P})$, $\forall x_j \in \mathcal{U}$ and $n^{\delta}_{\mathcal{Q}}(x_j) \in \mathcal{G}(\mathcal{Q})$. Then, the VPNCCR of the sample x_i within the positive region of the VPN

w.r.t. the subset of conditional attributes $\mathcal{Q}$ is defined as

$$\mathcal{K}(x_i) = \frac{|n_{\mathcal{P}}^{\delta}(x_i) \cap n_{\mathcal{Q}}^{\delta}(x_j)|}{|n_{\mathcal{P}}^{\delta}(x_i)|}. \tag{16}$$

$\mathcal{K}(x_i)$ measures the VPNCCR of any sample x_i within the positive region of the VPN w.r.t. the subset of conditional attributes $\mathcal{Q}$. It represents the proportion of the δ-neighborhood that contains samples of the same type, and $0 \le \mathcal{K}(x_i) \le 1$.

Based on Definition 13, the increase rate of the average VPNCCR considering conditional attributes will be defined.

Definition 14. Given a $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, for $\forall \mathcal{P}, \mathcal{Q} \subseteq \mathcal{C}$ and $\forall c_k \in \mathcal{C} - \mathcal{P}$. For $\forall x_i \in \mathcal{VPNPOS}^{\beta}_{\mathcal{P}\cup\{c_k\}}(\mathcal{Q})$, $\forall x_j \in \mathcal{VPNPOS}^{\beta}_{\mathcal{P}}(\mathcal{Q})$, $|\mathcal{VPNPOS}^{\beta}_{\mathcal{P}\cup\{c_k\}}(\mathcal{Q})| = m$, and $|\mathcal{VPNPOS}^{\beta}_{\mathcal{P}}(\mathcal{Q})| = n$, the increase rate of the average VPNCCR of c_k w.r.t. $\mathcal{P}$ considering $\mathcal{Q}$ is defined as

$$\mathcal{I}nc(c_k, \mathcal{P}, \mathcal{Q}) = \frac{\sum_{i=1}^{m} \mathcal{K}(x_i)}{m} - \frac{\sum_{i=1}^{n} \mathcal{K}(x_j)}{n}. \tag{17}$$

(1) When $\mathcal{I}nc(c_k, \mathcal{P}, \mathcal{Q}) < 0$, adding the attribute c_k that makes the quality degree of the positive region decrease.
(2) When $\mathcal{I}nc(c_k, \mathcal{P}, \mathcal{Q}) = 0$, adding the attribute c_k that makes the average gain flat and the quality degree of the positive region remains unchanged;
(3) When $\mathcal{I}nc(c_k, \mathcal{P}, \mathcal{Q}) > 0$, adding the attribute c_k that improves the quality degree of the positive region, which is the most desirable result.

Then, the $\mathcal{I}nc(c_k, \mathcal{P}, \mathcal{Q})$ can be used as the quality factor for the positive region, which can supervise the change of the positive region.

Definition 15. Given a $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, for $\forall \mathcal{P}, \mathcal{Q} \subseteq \mathcal{C}$ and $\forall c_k \in \mathcal{C} - \mathcal{P}$, the attribute quality degree of the attribute c_k w.r.t. $\mathcal{P}$ considering $\mathcal{Q}$ is defined as

$$\mathcal{A}qd(c_k, \mathcal{P}, \mathcal{Q}) = \mathcal{S}ig(c_k, \mathcal{P}, \mathcal{Q}) \times \mathcal{I}nc(c_k, \mathcal{P}, \mathcal{Q}). \tag{18}$$

$\mathcal{A}qd(c_k, \mathcal{P}, \mathcal{Q})$ is a function of the attribute itself c_k, the attribute c_k w.r.t. the subset of conditional attributes $\mathcal{P}$, and the subset of conditional attributes $\mathcal{Q}$.

The VPND $\gamma^{\beta}_{\mathcal{P}}(\mathcal{Q})$ mainly describes the relevance between the subset of conditional attributes $\mathcal{P}$ and $\mathcal{Q}$. To compute the dependency between the subset of conditional attributes, we should not try to check all subsets in the conditional attributes set $\mathcal{C}$. As there are $2^{|\mathcal{C}|}$ subsets in the conditional attributes set $\mathcal{C}$. Therefore, we will get $2^{2|\mathcal{C}|}$ dependencies. As the time complexity of computing dependencies of all conditional attributes subsets is exponential w.r.t. $|\mathcal{C}|$. Then, this is impractical. Therefore, we only consider dependencies w.r.t. all single-condition attribute subsets. $\mathcal{I}nc(c_k, \mathcal{P}, \mathcal{Q})$ mainly computes the increase rate of the average VPNCCR of the attribute c_k w.r.t. $\mathcal{P}$ considering $\mathcal{Q}$. Similarly, it is also impractical to compute the increase rate of the average VPNCCR w.r.t. all conditional attributes subsets. Therefore, we also consider only the VPNCCR w.r.t. all single-condition attribute subsets.

Definition 16. Given a $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, for $\forall \mathcal{P} \subseteq \mathcal{C}$ and $\forall c_k \in \mathcal{C} - \mathcal{P}$, the attribute quality degree of the attribute c_k w.r.t. $\mathcal{P}$ considering a subset of single-condition attribute $c_s (1 \le s \le m)$ is defined as

$$\mathcal{A}qd(c_k, \mathcal{P}, c_s) = \mathcal{S}ig(c_k, \mathcal{P}, c_s) \times \mathcal{I}nc(c_k, \mathcal{P}, c_s). \tag{19}$$

Then, the neighborhood quality degree of the attribute c_k w.r.t. $\mathcal{P}$ considering all single-condition attribute subsets is defined as

$$\mathcal{NA}qd(c_k, \mathcal{P}) = \frac{1}{m} \sum_{k=1}^{m} \mathcal{A}qd(c_k, \mathcal{P}, c_s). \tag{20}$$

Where $0 \le \mathcal{NA}qd(c_k, \mathcal{P}) \le 1$, in the same situation, its size can be used as an evaluation index for UAS.

Algorithm 1: The algorithm UAS_NAQD.

Input: $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, where $|\mathcal{C}| = m$, the neighborhood radius parameter δ, and the variable precision threshold parameter $\beta \in [0.5, 1.0]$, step size of 0.1.
Output: A reduced set $\mathcal{R}$.

```
R ← ∅, Label ← 1, B ← C − R;
for k ← 1 to m do
    Compute M(NR_{c_k}) by Eq. (7);
end
while Label do
    Let B = {c_{k_1}, c_{k_2}, ···, c_{k_h}};
    for l ← 1 to h do
        for s ← 1 to m do
            Compute γ^β_{R∪{c_{k_l}}}(c_s) by Eq. (14);
            Compute Aqd(c_{k_l}, R, c_s) by Eq. (18);
        end
        Compute NAqd(c_{k_l}, R) by Eq. (20);
        Compute Rel(R ∪ {c_{k_l}}) by Eq. (21);
    end
    Select the attribute c_{k_{l'}}, make NAqd(c_{k_{l'}}, R) have the
      maximum value;
    Compute SIG(c_{k_{l'}}, R) = Rel(R ∪ {c_{k_{l'}}}) − Rel(R) by
      Eq. (22);
    if SIG(c_{k_{l'}}, R) > 0 then
        R ← R ∪ {c_{k_{l'}}}, B ← B − {c_{k_{l'}}};
        B = {c_{k_1}, c_{k_2}, ···, c_{k_p}};
        for j ← 1 to p do
            Compute M(NR_R ∪ {c_{k_j}}) by Eq. (7);
        end
        else
            Label ← 0;
        end
    end
end
if |R| == m then
    return R ← R(1 : m − 1);
    else
        return R.
    end
end
```

Property 4. *Given $\forall \mathcal{Q} \subseteq \mathcal{C}, \delta(\delta \ge 0)$, and $\beta(0.5 \leqslant \beta \leqslant 1)$, for $\forall \mathcal{P}_1 \subseteq \mathcal{P}_2 \subseteq \mathcal{C}$, we have $\mathcal{VPNPOS}^{\beta}_{\mathcal{P}_1}(\mathcal{Q}) \subseteq \mathcal{VPNPOS}^{\beta}_{\mathcal{P}_2}(\mathcal{Q})$ and $\gamma^{\beta}_{\mathcal{P}_1}(\mathcal{Q}) \subseteq \gamma^{\beta}_{\mathcal{P}_2}(\mathcal{Q})$.*

Proof. Given $\forall \mathcal{Q} \subseteq \mathcal{C}$, for $\forall \mathcal{P}_1 \subseteq \mathcal{P}_2 \subseteq \mathcal{C}$, according to Property 1 and Property 3, we have $\mathcal{NR}_{\mathcal{P}_2} \subseteq \mathcal{NR}_{\mathcal{P}_1}$ and $n^{\delta}_{\mathcal{P}_2}(x_i) \subseteq n^{\delta}_{\mathcal{P}_1}(x_i)$. According to Definition 9 the lower approximation set of the VPN, we can get $\underline{\mathcal{NR}^{\beta}_{\mathcal{P}_1}}(n^{\delta}_{\mathcal{Q}}(x_i)) \subseteq \underline{\mathcal{NR}^{\beta}_{\mathcal{P}_2}}(n^{\delta}_{\mathcal{Q}}(x_i))$. So $\mathcal{VPNPOS}^{\beta}_{\mathcal{P}_1}(\mathcal{Q}) \subseteq \mathcal{VPNPOS}^{\beta}_{\mathcal{P}_2}(\mathcal{Q})$ and $\gamma^{\beta}_{\mathcal{P}_1}(\mathcal{Q}_2) \le \gamma^{\beta}_{\mathcal{P}_2}(\mathcal{Q}_1)$. □

Property 5. *Given $\forall \mathcal{P} \subseteq \mathcal{C}$, for $\forall \mathcal{Q}_1 \subseteq \mathcal{Q}_2 \subseteq \mathcal{C}$, $\delta(\delta \ge 0)$, and $\beta(0.5 \leqslant \beta \leqslant 1)$, we can get $\mathcal{VPNPOS}^{\beta}_{\mathcal{P}}(\mathcal{Q}_2) \subseteq \mathcal{VPNPOS}^{\beta}_{\mathcal{P}}(\mathcal{Q}_1)$ and $\gamma^{\beta}_{\mathcal{P}}(\mathcal{Q}_2) \le \gamma^{\beta}_{\mathcal{P}}(\mathcal{Q}_1)$.*

Proof. Given $\forall \mathcal{P} \subseteq \mathcal{C}$, for $\forall \mathcal{Q}_1 \subseteq \mathcal{Q}_2 \subseteq \mathcal{C}$, according to Property 1, we can get $n^{\delta}_{\mathcal{Q}_2}(x_i) \subseteq n^{\delta}_{\mathcal{Q}_1}(x_i)$. According to Definition 9 the lower approximation set of the VPN, we can get $\underline{\mathcal{NR}^{\beta}_{\mathcal{P}}}(n^{\delta}_{\mathcal{Q}_2}(x_i)) \subseteq \underline{\mathcal{NR}^{\beta}_{\mathcal{P}}}(n^{\delta}_{\mathcal{Q}_1}(x_i))$. So $\mathcal{VPNPOS}^{\beta}_{\mathcal{P}}(\mathcal{Q}_2) \subseteq \mathcal{VPNPOS}^{\beta}_{\mathcal{P}}(\mathcal{Q}_1)$ and $\gamma^{\beta}_{\mathcal{P}}(\mathcal{Q}_2) \le \gamma^{\beta}_{\mathcal{P}}(\mathcal{Q}_1)$. □

Property 6. *Given $\mathcal{Q} \subseteq \mathcal{C}$, for $\forall \mathcal{P}_1 \subseteq \mathcal{P}_2 \subseteq \cdots \subseteq \mathcal{P}_m \subseteq \mathcal{C}$, we can get $\gamma^{\beta}_{\mathcal{P}_1}(\mathcal{Q}) \le \gamma^{\beta}_{\mathcal{P}_2}(\mathcal{Q}) \le \cdots \le \gamma^{\beta}_{\mathcal{P}_m}(\mathcal{Q})$.*

Table 1
Initial and standardized VPNIS.

$\mathcal{U}$	c_1	c_2	c_3	c_4	c_5	c_6	c_1	c_2	c_3	c_4	c_5	c_6
x_1	1	0.3	90.86	C	a	A	0.0000	0.0000	0.8988	3	1	1
x_2	8	8.0	76.88	B	b	A	0.7778	0.8105	0.6668	2	2	1
x_3	4	5.6	88.69	B	b	A	0.3333	0.5579	0.8628	2	2	1
x_4	10	7.0	54.56	D	c	B	1.0000	0.7053	0.2965	4	3	2
x_5	5	1.8	96.96	A	c	B	0.4444	0.1579	1.0000	1	3	2
x_6	7	4.0	66.88	A	a	B	0.6667	0.3895	0.5009	1	1	2
x_7	9	9.8	96.23	C	d	C	0.8889	1.0000	0.9879	3	4	3
x_8	2	3.6	46.99	B	c	C	0.1111	0.3474	0.1709	2	3	3
x_9	7	0.6	36.69	D	a	C	0.6667	0.0316	0.0000	4	1	3

Property 7. *Given $\mathcal{P} \subseteq \mathcal{C}$, for $\forall \mathcal{Q}_1 \subseteq \mathcal{Q}_2 \subseteq \cdots \subseteq \mathcal{Q}_m \subseteq \mathcal{C}$, we can get $\gamma_{\mathcal{P}}^{\beta}(\mathcal{Q}_m) \leq \gamma_{\mathcal{P}}^{\beta}(\mathcal{Q}_{m-1}) \leq \cdots \leq \gamma_{\mathcal{P}}^{\beta}(\mathcal{Q}_1)$.*

The above Properties 4 - 7 describe some monotonicity properties of the VPND function w.r.t. the subset of conditional attributes. These properties are crucial for designing a forward search algorithm. Therefore, we adopt the VPND as a criterion for UAS. The algorithm search stop criterion is easy to perform.

Definition 17. The relevance of the subset of conditional attributes $\mathcal{B}$ is defined as

$$\mathcal{R}el(\mathcal{B}) = \frac{1}{m}\sum_{k=1}^{m}\gamma_{\mathcal{B}}^{\beta}(c_k). \tag{21}$$

Where $0 \leq \mathcal{R}el(P) \leq 1$, which reflects the relevance between the subset of conditional attributes $\mathcal{B}$ w.r.t. all single-condition attribute subsets.

Definition 18. Given $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, for $\forall \mathcal{B} \subseteq \mathcal{C}$ and $\forall b \in \mathcal{B}$, if $\mathcal{R}el(\mathcal{B}) = \mathcal{R}el(\mathcal{B} - b)$, we call b redundant in $\mathcal{B}$; otherwise, we call b necessary in $\mathcal{B}$. If every attribute in $\mathcal{B}$ is necessary, we call $\mathcal{B}$ independent.

Definition 19. Given $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, for $\forall \mathcal{B} \subseteq \mathcal{C}$ and $\forall b \in \mathcal{B}$, if $\mathcal{B}$ satisfies:

(1) $\mathcal{R}el(\mathcal{B}) = \mathcal{R}el(\mathcal{C})$;
(2) $\mathcal{R}el(\mathcal{B} - b) < \mathcal{R}el(\mathcal{B})$.

Then, we call $\mathcal{B}$ is a reduced set of $\mathcal{C}$.

In Definition 19, Condition (1) requires that attribute selection can not decrease the distinguishing ability of the system. The reduced set has the same distinguishing ability as the set of all conditional attributes; Condition (2) requires that there are no redundant attributes after attribute selection, all attributes are necessary. Definition 19 is completely consistent in form with the definition of supervised attribute selection in classical rough sets. However, we propose an unsupervised hybrid attribute selection model based on neighborhood attribute quality degree, which is a function of the attribute itself, the reduced set, and all single-condition attribute subsets, i.e., neighborhood attribute quality degree. Its size can be used as an evaluation index for UAS. At the same time, we use relevance to measure the correlation between conditional attributes. It can be used to define the attribute's significance. Supervised attribute selection uses the decision attribute to define the attribute's significance and the neighborhood attribute quality degree. Therefore, supervised attribute selection and UAS are completely different methods.

The relevance defines the correlation between conditional attributes. So, it can be used as an evaluation index of the attribute's significance. Next, the attribute's significance will be defined.

Definition 20. Give a $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, for $\forall \mathcal{B} \subseteq \mathcal{C}$ and $\forall c_k \in \mathcal{C} - \mathcal{B}$, the significance of the attribute c_k w.r.t. $\mathcal{B}$ considering all single-condition attribute subsets is defined as

$$\mathcal{SIG}(c_k, \mathcal{B}) = \mathcal{R}el(\mathcal{B} \cup \{c_k\}) - \mathcal{R}el(\mathcal{B}). \tag{22}$$

Where $0 \leq \mathcal{R}el(\mathcal{B}) \leq 1$ and $\mathcal{R}el(\mathcal{B}) \leq \mathcal{R}el(\mathcal{B} \cup \{c_k\})$, we can get $0 \leq \mathcal{SIG}(c_k, \mathcal{P}) \leq 1$. If $\mathcal{SIG}(c_k, \mathcal{P}) = 0$, we call c_k redundant in $\mathcal{B}$. Otherwise, we call c_k necessary in $\mathcal{B}$.

4.2. Neighborhood attribute quality degree-based unsupervised hybrid attribute selection algorithm

In this section, the neighborhood attribute quality degree-based unsupervised hybrid attribute selection model will be proposed, and the corresponding algorithm UAS_NAQD will be analyzed specifically.

In real life, there is usually a large difference in how people describe data samples. To obtain accurate data processing results, we need to standardize the numerical attribute data in the original data. In this paper, we adopt the computational method of Refs. Zhang et al. (2022a, 2024).

Next, UAS_NAQD will be designed, and its time complexity will be analyzed.

Algorithm 1 starts with an empty set $\mathcal{R}$. First, the neighborhood attribute quality degree and significance of all remaining attributes are computed. Then, the attribute with the largest neighborhood quality degree is selected to add to $\mathcal{R}$ until the significance of all remaining attributes is 0, i.e., the significance of the system does not change with the addition of any new attribute. In Algorithm 1, the number of cycles in Steps 2–4 is m; the number of cycles in Step 3 is $n \times n$; the number of cycles in Steps 7–12 is $h \times m$; the number of cycles in Steps 8–10 is m; the number of cycles in Steps 18–20 is p; the number of cycles in Step 19 is $n \times n$. Then, the total number of cycles of Algorithm 1 is $m \times n \times n + h \times m + p \times n \times n$. So, the time complexity of Algorithm 1 is $O((m+p)n^2)$.

4.3. A special example

In this section, a VPNIS example that explains the above study is shown.

Give a $VPNIS = (\mathcal{U}, \mathcal{C}, \mathcal{V}, f, \delta, \beta)$, as shown in the left side of Table 1, which is mainly hybrid attribute data, where $\mathcal{U} = \{x_1, x_2, \cdots, x_9\}$ and $\mathcal{C} = \{c_1, c_2, \cdots, c_6\}$. Among these attributes, c_1, c_2, and c_3 are numerical attributes; c_4, c_5, and c_6 are nominal attributes. The right side of Table 1 shows the standardized result. Let $\mathcal{R} = \emptyset$, $\mathcal{B} = \mathcal{C}$, and $\beta = 0.6$.

For $\forall c_s \in \mathcal{C}$, its neighborhood relation matrix can be computed, which is as follows.

Table 2
The VPND of $\mathcal{R} \cup \{c_k\}$ w.r.t. c_s.

$\gamma^{0.6}_{\mathcal{R}\cup\{c_k\}}(c_s)$	c_1	c_2	c_3	c_4	c_5	c_6
$\mathcal{R} \cup \{c_1\}$	1.0000	1.0000	0.8889	0.2222	0.1111	0.1111
$\mathcal{R} \cup \{c_2\}$	0.8889	1.0000	0.8889	0.1111	0.3333	0.1111
$\mathcal{R} \cup \{c_3\}$	1.0000	1.0000	1.0000	0.1111	0.1111	0.5556
$\mathcal{R} \cup \{c_4\}$	0.7778	0.5556	1.0000	1.0000	0.3333	0.5556
$\mathcal{R} \cup \{c_5\}$	0.7778	1.0000	1.0000	0.3333	1.0000	0.6667
$\mathcal{R} \cup \{c_6\}$	1.0000	1.0000	1.0000	0.6667	0.6667	1.0000

Table 3
The significance of c_k w.r.t. $\mathcal{R}$ considering c_s.

$Sig(c_k, \mathcal{R}, c_s)$	c_1	c_2	c_3	c_4	c_5	c_6
c_1	1.0000	1.0000	0.8889	0.2222	0.1111	0.1111
c_2	0.8889	1.0000	0.8889	0.1111	0.3333	0.1111
c_3	1.0000	1.0000	1.0000	0.1111	0.1111	0.5556
c_4	0.7778	0.5556	1.0000	1.0000	0.3333	0.5556
c_5	0.7778	1.0000	1.0000	0.3333	1.0000	0.6667
c_6	1.0000	1.0000	1.0000	0.6667	0.6667	1.0000

$$\mathcal{M}(\mathcal{NR}_{c_1}) = \begin{pmatrix} 1&0&0&0&0&0&0&1&0\\ 0&1&0&1&0&1&1&0&1\\ 0&0&1&0&1&0&0&1&0\\ 0&1&0&1&0&0&1&0&0\\ 0&0&1&0&1&1&0&0&1\\ 0&1&0&0&1&1&1&0&1\\ 0&1&0&1&0&1&1&0&1\\ 1&0&1&0&0&0&0&1&0\\ 0&1&0&0&1&1&1&0&1 \end{pmatrix}, \mathcal{M}(\mathcal{NR}_{c_2}) = \begin{pmatrix} 1&0&0&0&1&0&0&0&1\\ 0&1&1&1&0&0&1&0&0\\ 0&1&1&1&0&1&0&1&0\\ 0&1&1&1&0&0&0&0&0\\ 1&0&0&0&1&1&0&1&1\\ 0&0&1&0&1&1&0&1&0\\ 0&1&0&0&0&0&1&0&0\\ 0&0&1&0&1&1&0&1&0\\ 1&0&0&0&1&0&0&0&1 \end{pmatrix},$$

$$\mathcal{M}(\mathcal{NR}_{c_3}) = \begin{pmatrix} 1&1&1&0&1&0&1&0&0\\ 1&1&1&0&0&1&0&0&0\\ 1&1&1&0&1&0&1&0&0\\ 0&0&0&1&0&1&0&1&0\\ 1&0&1&0&1&0&1&0&0\\ 0&1&0&1&0&1&0&0&0\\ 1&0&1&0&1&0&1&0&0\\ 0&0&0&1&0&0&0&1&1\\ 0&0&0&0&0&0&0&1&1 \end{pmatrix}, \mathcal{M}(\mathcal{NR}_{c_4}) = \begin{pmatrix} 1&0&0&0&0&0&1&0&0\\ 0&1&1&0&0&0&0&1&0\\ 0&1&1&0&0&0&0&1&0\\ 0&0&0&1&0&0&0&0&1\\ 0&0&0&0&1&1&0&0&0\\ 0&0&0&0&1&1&0&0&0\\ 1&0&0&0&0&0&1&0&0\\ 0&1&1&0&0&0&0&1&0\\ 0&0&0&1&0&0&0&0&1 \end{pmatrix},$$

$$\mathcal{M}(\mathcal{NR}_{c_5}) = \begin{pmatrix} 1&0&0&0&0&1&0&0&1\\ 0&1&1&0&0&0&0&0&0\\ 0&1&1&0&0&0&0&0&0\\ 0&0&0&1&1&0&0&1&0\\ 0&0&0&1&1&0&0&1&0\\ 1&0&0&0&0&1&0&0&1\\ 0&0&0&0&0&0&1&0&0\\ 0&0&0&1&1&0&0&1&0\\ 1&0&0&0&0&1&0&0&1 \end{pmatrix}, \mathcal{M}(\mathcal{NR}_{c_6}) = \begin{pmatrix} 1&1&1&0&0&0&0&0&0\\ 1&1&1&0&0&0&0&0&0\\ 1&1&1&0&0&0&0&0&0\\ 0&0&0&1&1&1&0&0&0\\ 0&0&0&1&1&1&0&0&0\\ 0&0&0&1&1&1&0&0&0\\ 0&0&0&0&0&0&1&1&1\\ 0&0&0&0&0&0&1&1&1\\ 0&0&0&0&0&0&1&1&1 \end{pmatrix}.$$

Table 4
The increase rate of the average VPNCCR of c_k w.r.t. $\mathcal{R}$ considering c_s.

$\mathcal{I}nc(c_k, \mathcal{R}, c_s)$	c_1	c_2	c_3	c_4	c_5	c_6
c_1	1.0000	0.8296	0.6937	0.6667	0.6667	0.6667
c_2	0.7271	1.0000	0.6854	0.6000	0.6222	0.6000
c_3	0.6648	0.6926	1.0000	0.6667	0.6667	0.6133
c_4	0.8571	1.0000	0.8889	1.0000	0.6667	0.8000
c_5	0.7143	1.0000	0.7778	1.0000	1.0000	0.8333
c_6	0.6667	0.7778	0.7778	0.6667	0.6667	1.0000

Table 5
The attribute quality degree of c_k w.r.t. $\mathcal{R}$ considering c_s.

$\mathcal{A}qd(c_k, \mathcal{R}, c_s)$	c_1	c_2	c_3	c_4	c_5	c_6
c_1	1.0000	0.8296	0.6167	0.1481	0.0741	0.0741
c_2	0.6463	1.0000	0.6093	0.0667	0.2074	0.0667
c_3	0.6648	0.6926	1.0000	0.0741	0.0741	0.3407
c_4	0.6667	0.5556	0.8889	1.0000	0.2222	0.4444
c_5	0.5556	1.0000	0.7778	0.3333	1.0000	0.5556
c_6	0.6667	0.7778	0.7778	0.4444	0.4444	1.0000

By Definition 11, for $\forall c_k \in \mathcal{B}$ and $\forall c_s \in \mathcal{C}$, the VPND of $\mathcal{R} \cup \{c_k\}$ w.r.t. the subset of single-condition attribute c_s can be computed, which is shown in Table 2.

By Definition 12, the significance of the attribute c_k w.r.t. $\mathcal{R}$ considering the subset of single-condition attribute c_s can be computed, which is shown in Table 3.

By Definition 14, the increase rate of the average VPNCCR of the attribute c_k w.r.t. $\mathcal{R}$ considering the subset of single-condition attribute c_s can be computed, which is shown in Table 4.

By Definition 15, the attribute quality degree of the attribute c_k w.r.t. $\mathcal{R}$ considering the subset of single-condition attribute c_s can be computed, which is shown in Table 5.

Then, from Eq. 20, the neighborhood attribute quality degree of the attribute c_k w.r.t. $\mathcal{R}$ can be computed, which is as follows. $\mathcal{NA}qd(c_1, \mathcal{R}) = \frac{1}{6}(1.0000 + 0.8296 + 0.6167 + 0.1481 + 0.0741 + 0.0741) \approx 0.4571$; $\mathcal{NA}qd(c_2, \mathcal{R}) \approx 0.4327$; $\mathcal{NA}qd(c_3, \mathcal{R}) \approx 0.4744$; $\mathcal{NA}qd(c_4, \mathcal{R}) \approx 0.6296$; $\mathcal{NA}qd(c_5, \mathcal{R}) \approx 0.7037$; $\mathcal{NA}qd(c_6, \mathcal{R}) \approx 0.6852$.

Next, the attribute c_5 with the largest neighborhood attribute quality degree can be selected.

By Definition 17, the relevance of $\mathcal{R} \cup \{c_5\}$ can be computed (At this point $\mathcal{R} = \emptyset$, therefore $\mathcal{R}el(\mathcal{R}) = 0.0000$). $\mathcal{R}el(\mathcal{R} \cup \{c_5\}) = \frac{1}{6}(0.7778 + 1.0000 + 1.0000 + 0.3333 + 1.0000 + 0.6667) \approx 0.7963$;

By Definition 20, the significance of c_5 w.r.t. $\mathcal{R}$ can be computed. $\mathcal{SIG}(c_5, \mathcal{R}) = \mathcal{R}el(\mathcal{R} \cup \{c_5\}) - \mathcal{R}el(\mathcal{R}) = 0.7963 - 0.0000 = 0.7963$.

Therefore, the attribute c_5 can be added to $\mathcal{R}$. UAS_NAQD can get $\mathcal{R} = \{c_5\}$ and $\mathcal{B} = \mathcal{C} - \mathcal{R} = \{c_1, c_2, c_3, c_4, c_6\}$.

Similarly, UAS_NAQD can compute $\mathcal{NA}qd(c_1, \mathcal{R}) = 0.0137$; $\mathcal{NA}qd(c_2, \mathcal{R}) = -0.0301$; $\mathcal{NA}qd(c_3, \mathcal{R}) = -0.0075$; $\mathcal{NA}qd(c_4, \mathcal{R}) = 0.0093$; $\mathcal{NA}qd(c_6, \mathcal{R}) = -0.0095$. The attribute c_1 with the largest neighorhood attribute quality degree can be selected. $\mathcal{R}el(\mathcal{R} \cup \{c_1\}) = 0.9259$, $\mathcal{SIG}(c_1, \mathcal{R}) = \mathcal{R}el(\mathcal{R} \cup \{c_1\}) - \mathcal{R}el(\mathcal{R}) = 0.9529 - 0.7963 = 0.1566$. Therefore, the attribute c_1 can be added to $\mathcal{R}$. UAS_NAQD can get $\mathcal{R} = \{c_5, c_1\}$ and $\mathcal{B} = \mathcal{C} - \mathcal{R} = \{c_2, c_3, c_4, c_6\}$.

Similarly, UAS_NAQD can compute $\mathcal{NA}qd(c_2, \mathcal{R}) = 0.7407 \times 10^{-5}$; $\mathcal{NA}qd(c_3, \mathcal{R}) = 0.7407 \times 10^{-5}$; $\mathcal{NA}qd(c_4, \mathcal{R}) = 0.7407 \times 10^{-5}$; $\mathcal{NA}qd(c_6, \mathcal{R}) = 0.7407 \times 10^{-5}$. The attribute c_2 with the largest neighborhood attribute quality degree can be selected. $\mathcal{R}el(\mathcal{R} \cup c_2) = 1.0000$, $\mathcal{SIG}(c_2, \mathcal{R}) = \mathcal{R}el(\mathcal{R} \cup c_2) - \mathcal{R}el(\mathcal{R}) = 1.0000 - 0.9529 = 0.0471$. Therefore, the attribute c_2 can be added to $\mathcal{R}$. UAS_NAQD can get $\mathcal{R} = \{c_5, c_1, c_2\}$,$\mathcal{B} = \mathcal{C} - \mathcal{R} = \{c_3, c_4, c_6\}$.

Similarly, UAS_NAQD can compute $\mathcal{NA}qd(c_3, \mathcal{R}) = 0.0000$; $\mathcal{NA}qd(c_4, \mathcal{R}) = 0.0000$; $\mathcal{NA}qd(c_6, \mathcal{R}) = 0.0000$. The attribute c_3 with the largest neighborhood attribute quality degree can be selected. $\mathcal{R}el(\mathcal{R} \cup c_3) = 1.0000$, $\mathcal{SIG}(c_3, \mathcal{R}) = \mathcal{R}el(\mathcal{R} \cup c_2) - \mathcal{R}el(\mathcal{R}) = 1.0000 - 1.0000 = 0.0000$.

Therefore, $\mathcal{R} = \{c_5, c_1, c_2\}$.

5. Experiments and analysis

In this section, the performance of UAS_NAQD is evaluated by clustering tasks. It mainly includes experimental datasets, experimental preparations, experimental results, parameter sensitivity analysis, and statistical tests.

5.1. Experimental datasets

The 32 datasets from publicly available databases are chosen for experiments, which include numerical (12), nominal (10), and hybrid (10) attribute datasets. For missing values in certain datasets, we use the maximum probability value method to impute the missing values, i.e., to fill the missing attribute values with the value that appears most frequently in other samples on this attribute (Hu et al., 2025; Yang et al., 2024). The descriptions of datasets used for clustering experiments are shown in Table 6.

5.2. Experimental preparations

The experimental datasets are given in Table 6. Meanwhile, comparison algorithms are presented in Fig. 2. In this section, UAS_NAQD is compared with existing UAS algorithms at this stage. Comparison algorithms mainly include the Laplacian Score for feature selection (LS) (He et al., 2005), the Unsupervised Feature Selection using Feature Similarity (UFSFS) (Mitra et al., 2002), the Unsupervised Spectral Feature Selection Method (USFSM) (Solorio-Fernández et al., 2017), the SPECtral analysis-based (SPEC) (Zhao & Liu, 2007), the Unsupervised fuzzy-rough set-based dimensionality reduction (UFRFS) (Mac Parthaláin & Jensen, 2013), the Fuzzy Rough set-based Unsupervised Attribute Reduction (FRUAR) (Yuan et al., 2021a), the Hybrid-Kernel-based fuzzy Complementary Mutual Information (HKCMI) (Yuan et al., 2021b), the Fuzzy Mutual Information-based Unsupervised Feature Selection (FMIUFS) (Yuan et al., 2021c), and the Multi-scale Information Fusion-based Multiple Correlations for Unsupervised Attribute Selection (UAS_MsIFMC) (Zhang et al., 2024). The detailed descriptions of UAS comparison algorithms are shown in Table 7.

In Table 7, the "-" means that the algorithm has no parameter. For algorithm FRUAR, which adjusts the parameter λ from 0.1 to 3.0 with a step size of 0.1; For algorithm HKCMI, which adjusts the parameter ρ from 0.1 to 1.0 with a step size of 0.1, and another parameter $\delta \in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$; for algorithm FMIUFS, which

Table 6
The descriptions of datasets.

No.	Dataset	Samples	Conditional attributes		Decision classes
			Numerical	Nominal	
D01	Diabetes	768	8	0	2
D02	Ecoli	336	7	0	8
D03	Glass Identification	214	10	0	6
D04	Ionosphere	351	33	0	2
D05	Libras Movement	360	90	0	15
D06	Parkinsons	195	22	0	2
D07	Sonar	208	60	0	2
D08	Wisconsin Breast Cancer	699	9	0	2
D09	Wisconsin Diagnostic Breast Cancer	569	31	0	2
D10	Wilt	4839	5	0	2
D11	winequality_white	4898	11	0	7
D12	Wisconsin Prognostic Breast Cancer	198	34	0	2
D13	Audiology	226	0	69	24
D14	Balance Scale	625	0	4	3
D15	Car	1728	0	6	4
D16	Chess	3196	0	36	2
D17	Lung Cancer	32	0	56	4
D18	Lymphography	148	0	18	4
D19	Monks	432	0	6	2
D20	Soybean(Large)	683	0	35	19
D21	Soybean(Small)	47	0	35	4
D22	Spect	267	0	22	2
D23	Autos	205	15	10	6
D24	Bands	531	24	15	2
D25	Credit Approval	690	6	9	2
D26	Flag	194	9	19	8
D27	German	1000	7	13	2
D28	Heart	270	6	7	2
D29	Hepatitis	155	6	13	2
D30	Horse	368	7	20	2
D31	Hypothyroid	3163	7	18	2
D32	Sick	3772	6	23	2

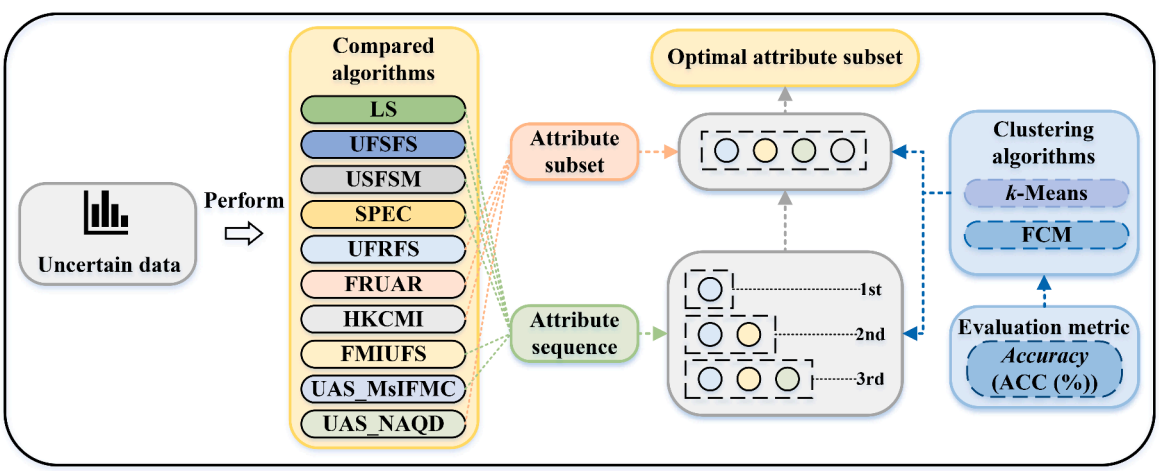
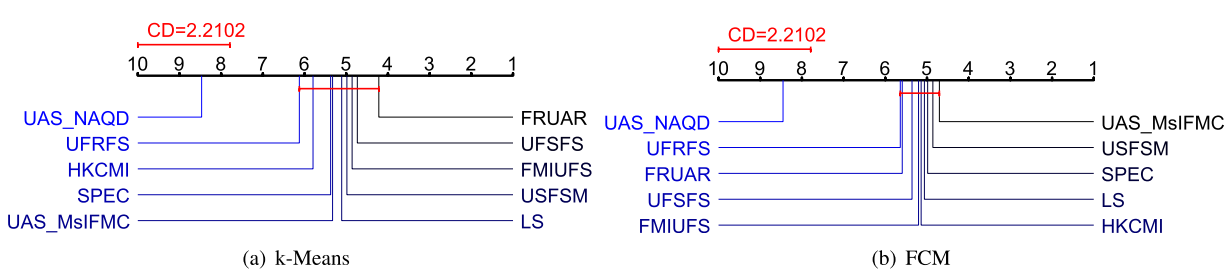
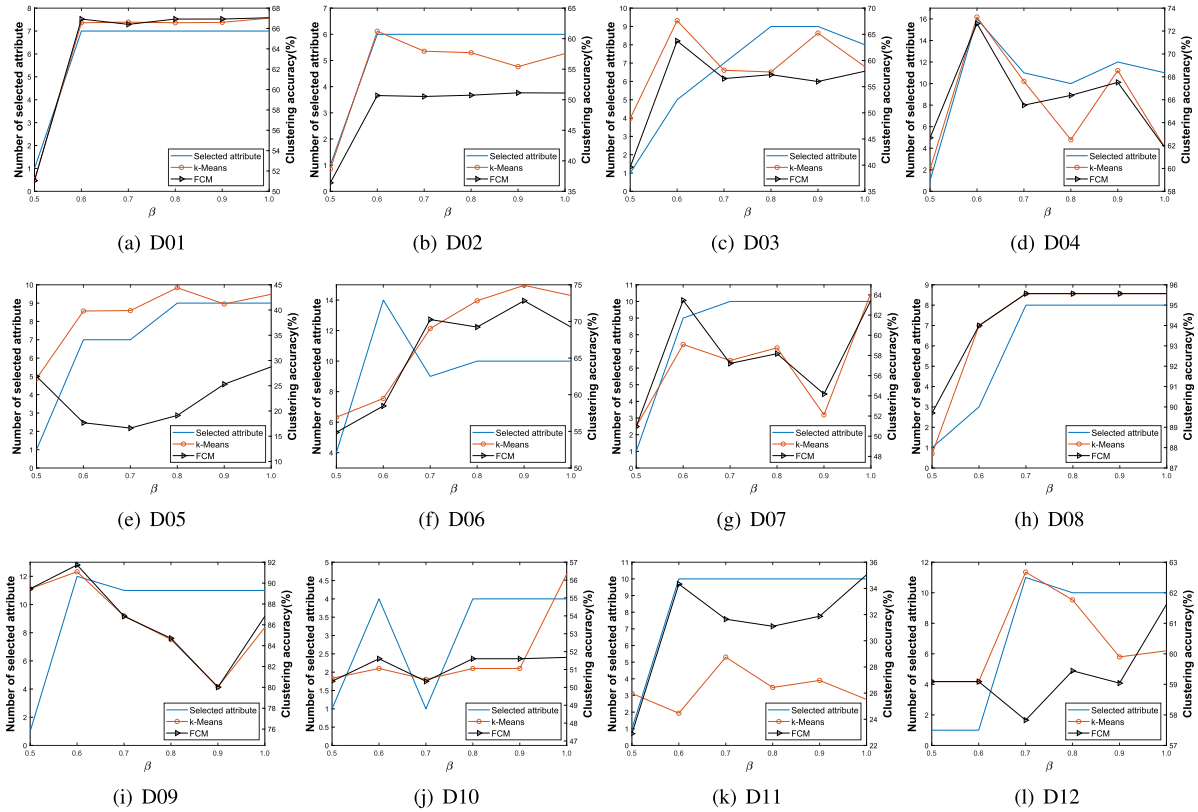


Fig. 2. The framework for evaluating UAS algorithms.



(a) k-Means (b) FCM

Fig. 3. Nemenyi's test figures are based on the clustering algorithms.

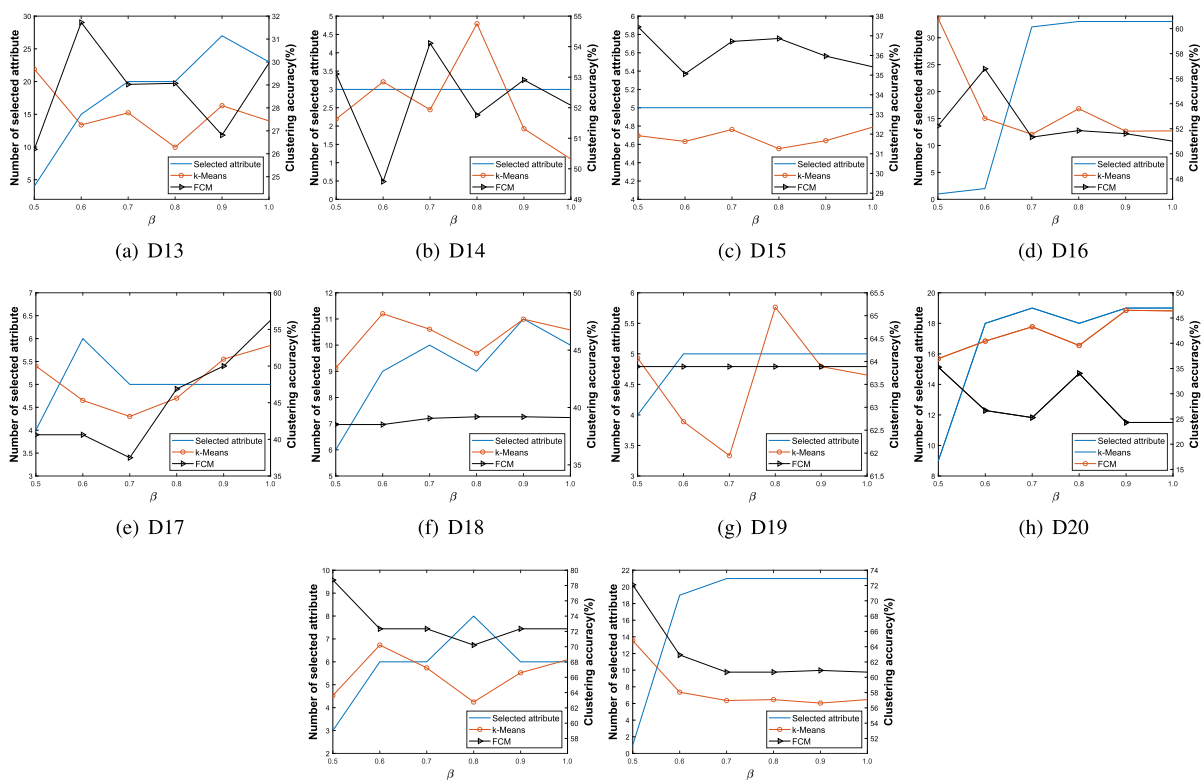


(a) D01 (b) D02 (c) D03 (d) D04

(e) D05 (f) D06 (g) D07 (h) D08

(i) D09 (j) D10 (k) D11 (l) D12

Fig. 4. The number of selected attributes and ACC on numerical attribute datasets using different β values.



(a) D13 (b) D14 (c) D15 (d) D16

(e) D17 (f) D18 (g) D19 (h) D20

(i) D21 (j) D22

Fig. 5. The number of selected attributes and ACC on nominal attribute datasets using different β values.

Table 7
The detailed descriptions of UAS comparison algorithms.

Comparison algorithm	Main idea	Application scope	parameter
LS He et al. (2005)	Using Laplace Scores to evaluate the attribute's significance to rank attribute	Numerical	-
UFSFS Mitra et al. (2002)	Introduction of maximum information compression index to rank attribute	Numerical	-
USFSM Solorio-Fernández et al. (2017)	Kernel and spectral-based attribute evaluation measures to rank attribute	hybrid	-
SPEC Zhao and Liu (2007)	A unified framework based on spectral graph theory to rank attribute	Numerical	-
UFRFS Mac Parthaláin and Jensen (2013)	The fuzzy-rough sets-based to attribute selection	hybrid	-
FRUAR Yuan et al. (2021a)	The fuzzy dependency-based to attribute selection	hybrid	$\lambda \in [0.1, 3.0]$
HKCMI Yuan et al. (2021b)	The hybrid-kernel fuzzy complementary entropy-based to attribute selection	hybrid	$\rho \in [0.1, 1.0]$ and $\delta \in \{10^{-2}, 10^{-3}, 10^{-4}, 10^{-5}\}$
FMIUFS Yuan et al. (2021c)	The fuzzy mutual information-based to rank attribute	hybrid	$\lambda \in [0.1, 2.0]$
UAS_MsIFMC Zhang et al. (2024)	The multi-scale information fusion-based to rank attribute	hybrid	$\lambda \in \{0.1, 0.5, 1.0\}$
UAS_NAQD	The neighborhood attribute quality degree to attribute selection	hybrid	$\beta \in [0.5, 1.0]$

Table 8
The optimal number of attributes for the clustering algorithm k-Means.

Dataset	Original attributes	LS	UFSFS	USFSM	SPEC	UFRFS	FRUAR	HKCMI	FMIUFS	UAS_MsIFMC	UAS_NAQD
D01	8.0	7.0	7.0	7.0	7.0	**8.0**	7.0	7.0	7.0	7.0	7.0
D02	7.0	6.0	6.0	6.0	6.0	**7.0**	6.0	6.0	6.0	6.0	6.0
D03	10.0	5.0	5.0	5.0	5.0	8.0	9.0	9.0	5.0	5.0	5.0
D04	33.0	16.0	16.0	16.0	16.0	12.0	30.0	29.0	16.0	16.0	16.0
D05	90.0	9.0	9.0	9.0	9.0	5.0	35.0	34.0	9.0	9.0	9.0
D06	22.0	10.0	10.0	10.0	10.0	8.0	15.0	21.0	10.0	10.0	10.0
D07	60.0	10.0	10.0	10.0	10.0	8.0	19.0	30.0	10.0	10.0	10.0
D08	9.0	8.0	8.0	8.0	8.0	**9.0**	8.0	8.0	8.0	8.0	8.0
D09	31.0	12.0	12.0	12.0	12.0	9.0	24.0	29.0	12.0	12.0	12.0
D10	5.0	4.0	4.0	4.0	4.0	5.0	4.0	4.0	4.0	4.0	4.0
D11	11.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
D12	34.0	11.0	11.0	11.0	11.0	8.0	12.0	33.0	11.0	11.0	11.0
D13	69.0	4.0	4.0	4.0	4.0	25.0	23.0	8.0	4.0	4.0	4.0
D14	4.0	3.0	3.0	3.0	3.0	**4.0**	3.0	3.0	3.0	3.0	3.0
D15	6.0	5.0	5.0	5.0	5.0	**6.0**	5.0	5.0	5.0	5.0	5.0
D16	36.0	1.0	1.0	1.0	1.0	32.0	33.0	11.0	1.0	1.0	1.0
D17	56.0	5.0	5.0	5.0	5.0	43.0	5.0	5.0	5.0	5.0	5.0
D18	18.0	9.0	9.0	9.0	9.0	11.0	10.0	7.0	9.0	9.0	9.0
D19	6.0	5.0	5.0	5.0	5.0	**6.0**	5.0	5.0	5.0	5.0	5.0
D20	35.0	19.0	19.0	19.0	19.0	25.0	19.0	6.0	19.0	19.0	19.0
D21	35.0	6.0	6.0	6.0	6.0	10.0	6.0	5.0	6.0	6.0	6.0
D22	22.0	1.0	1.0	1.0	1.0	20.0	21.0	12.0	1.0	1.0	1.0
D23	25.0	12.0	12.0	12.0	12.0	17.0	14.0	5.0	12.0	12.0	12.0
D24	39.0	2.0	2.0	2.0	2.0	18.0	7.0	1.0	2.0	2.0	2.0
D25	15.0	14.0	14.0	14.0	14.0	**15.0**	14.0	7.0	14.0	14.0	14.0
D26	28.0	14.0	14.0	14.0	14.0	20.0	13.0	7.0	14.0	14.0	14.0
D27	20.0	9.0	9.0	9.0	9.0	19.0	14.0	6.0	9.0	9.0	9.0
D28	13.0	5.0	5.0	5.0	5.0	**13.0**	12.0	8.0	5.0	5.0	5.0
D29	19.0	15.0	15.0	15.0	15.0	18.0	14.0	11.0	15.0	15.0	15.0
D30	27.0	5.0	5.0	5.0	5.0	20.0	5.0	1.0	5.0	5.0	5.0
D31	25.0	2.0	2.0	2.0	2.0	23.0	23.0	15.0	2.0	2.0	2.0
D32	29.0	25.0	25.0	25.0	25.0	27.0	25.0	11.0	25.0	25.0	25.0
Avg.	26.5	8.4	8.4	8.4	8.4	14.7	14.1	11.2	8.4	8.4	8.4

adjusts the parameter λ from 0.1 to 2.0 with a step size of 0.1; for algorithm UAS_MsIFMC, which adjusts the parameter $\lambda \in \{0.1, 0.5, 1.0\}$. They choose an optimal subset of attributes for each dataset.

In clustering experiments, the clustering algorithms k-Means and Fuzzy C-Means (FCM) are used to evaluate the performance of comparison algorithms. They return predictive labels of the dataset. We use the clustering accuracy (ACC) Zhang et al. (2023b, 2024) to evaluate the performance of clustering algorithms. The larger ACC of the clustering algorithms indicates their better performance. Since the k-Means and FCM are random, so we repeat the experiment 10 times. Finally, this paper uses the average and standard deviation of ACC as the final result.

In the experiment, we first standardize the numerical attribute data in the dataset to [0,1]. Then, in Fig. 2, we follow the preferred experimental setup in Ref. Yuan et al. (2021c) when the algorithm LS computes the neighbor graph matrix, its neighborhood size is set to a fixed value of 5. For algorithm UFSFS, its “style" is set to -1.

For algorithms LS, UFSFS, USFSM, SPEC, FMIUFS, and UAS_MsIFMC, their output is a sequence of attributes arranged in descending order of the attribute's significance. To compare with UAS_NAQD, we choose

Table 9
The optimal number of attributes for the clustering algorithm FCM.

Dataset	Original attributes	LS	UFSFS	USFSM	SPEC	UFRFS	FRUAR	HKCMI	FMIUFS	UAS_MsIFMC	UAS_NAQD
D01	8	7	7	7	7	**8**	7	7	7	7	7
D02	7	6	6	6	6	**7**	6	6	6	6	6
D03	10	5	5	5	5	8	9	9	5	5	5
D04	33	16	16	16	16	12	32	12	16	16	16
D05	90	9	9	9	9	5	49	8	9	9	9
D06	22	10	10	10	10	8	18	8	10	10	10
D07	60	9	9	9	9	8	30	8	9	9	9
D08	9	8	8	8	8	**9**	8	8	8	8	8
D09	31	12	12	12	12	9	29	12	12	12	12
D10	5	4	4	4	4	5	4	4	4	4	4
D11	11	10	10	10	10	10	10	10	10	10	10
D12	34	10	10	10	10	8	24	9	10	10	10
D13	69	15	15	15	15	25	23	8	15	15	15
D14	4	3	3	3	3	**4**	3	3	3	3	3
D15	6	5	5	5	5	**6**	5	5	5	5	5
D16	36	2	2	2	2	32	33	11	2	2	2
D17	56	5	5	5	5	43	5	5	5	5	5
D18	18	9	9	9	9	11	10	7	9	9	9
D19	6	4	4	4	4	**6**	5	5	4	4	4
D20	35	9	9	9	9	25	19	6	9	9	9
D21	35	3	3	3	3	10	6	5	3	3	3
D22	22	1	1	1	1	20	21	12	1	1	1
D23	25	3	3	3	3	17	16	7	3	3	3
D24	39	7	7	7	7	18	9	2	7	7	7
D25	15	14	14	14	14	**15**	14	8	14	14	14
D26	28	6	6	6	6	20	17	7	6	6	6
D27	20	15	15	15	15	19	17	7	15	15	15
D28	13	5	5	5	5	**13**	12	9	5	5	5
D29	19	15	15	15	15	18	15	10	15	15	15
D30	27	1	1	1	1	20	5	1	1	1	1
D31	25	2	2	2	2	23	23	12	2	2	2
D32	29	1	1	1	1	27	26	13	1	1	1
Avg.	26.5	7.2	7.2	7.2	7.2	14.7	15.9	7.6	7.2	7.2	7.2

Table 10
The attribute selection data on the clustering algorithm k-Means ACC(%).

Dataset	Original attributes	LS	UFSFS	USFSM	SPEC	UFRFS	FRUAR	HKCMI	FMIUFS	UAS_MsIFMC	UAS_NAQD	β
D01	66.24±1.21	66.34±2.22	66.51±1.58	66.78±2.33	66.94±1.90	66.46±1.07	66.60±0.07	66.67±0.11	65.94±0.83	65.55±0.76	**67.02±1.83**	1.0
D02	55.42±5.35	58.27±9.24	58.72±9.00	58.48±5.51	55.45±4.44	59.70±6.39	56.25±5.12	60.54±12.22	60.77±7.93	59.82±7.63	**61.22±4.84**	0.6
D03	58.36±6.99	66.96±7.01	66.87±7.64	65.75±6.93	68.18±7.55	58.41±8.02	56.31±8.55	65.28±10.09	66.92±7.61	**69.39±4.86**	67.62±8.11	0.6
D04	71.08±0.15	70.71±0.44	70.46±0.27	70.54±0.31	70.71±0.32	68.69±2.42	69.37±0.24	70.94±0.00	70.63±0.28	70.43±0.12	**73.22±0.00**	0.6
D05	43.83±2.38	32.03±1.98	31.86±1.46	32.28±0.98	32.25±0.93	31.33±1.09	42.78±2.66	42.86±2.36	32.39±1.48	31.53±0.90	**44.42±2.39**	0.8
D06	62.82±0.27	68.87±4.44	70.36±4.62	71.44±3.75	68.46±5.28	74.87±1.28	66.15±0.00	62.87±0.26	66.67±5.30	69.44±6.33	**74.92±1.99**	0.9
D07	53.37±1.86	58.08±1.45	58.32±1.43	57.45±1.04	58.46±1.49	56.25±0.00	55.96±1.25	54.23±0.63	57.36±0.96	58.08±1.22	**64.18±1.86**	1.0
D08	95.85±0.00	**95.99±0.00**	**95.99±0.00**	**95.99±0.00**	**95.99±0.00**	95.85±0.00	95.57±0.00	**95.99±0.00**	**95.99±0.00**	**95.99±0.00**	95.57±0.00	0.7
D09	92.79±0.00	90.12±0.53	87.63±8.83	90.32±0.56	82.27±13.61	91.42±0.07	85.31±11.99	**93.08±0.15**	90.39±0.63	90.60±0.59	91.09±0.08	0.6
D10	51.15±0.20	**63.01±19.57**	50.86±0.03	54.90±12.81	54.91±12.81	55.24±12.88	51.11±0.08	51.06±0.01	54.91±12.81	54.90±12.81	56.34±13.44	1.0
D11	26.49±2.35	26.02±2.07	24.89±2.86	24.93±2.66	25.15±1.94	26.15±1.55	28.07±1.79	26.47±1.55	26.11±1.76	25.49±1.18	**28.75±2.56**	0.7
D12	59.80±0.26	60.86±3.53	61.67±4.21	61.87±4.06	60.20±2.73	60.66±9.49	54.55±0.00	59.80±0.26	58.59±2.96	59.14±0.77	**62.68±5.46**	0.7
D13	29.07±2.53	41.19±0.14	41.15±0.00	**41.24±0.28**	41.19±0.14	29.34±2.39	26.24±2.15	28.01±1.92	41.15±0.00	41.15±0.00	29.69±0.61	0.5
D14	50.98±2.33	51.44±4.00	52.64±3.64	51.41±3.77	54.30±3.51	52.19±3.64	49.92±5.82	52.21±5.35	51.30±1.58	51.65±4.58	**54.75±2.58**	0.8
D15	32.22±1.60	32.11±1.67	31.30±2.14	**33.00±2.74**	32.00±2.06	31.49±1.98	30.86±1.90	31.73±2.88	31.99±1.49	31.72±2.35	32.36±1.99	1.0
D16	52.96±3.73	51.69±0.00	51.69±0.00	51.69±0.00	51.69±0.00	51.92±3.01	50.97±0.00	55.70±3.84	51.69±0.00	51.69±0.00	**60.86±0.00**	0.5
D17	55.94±7.86	47.81±5.11	46.88±4.17	44.38±1.98	47.81±5.32	53.75±5.86	**58.13±3.67**	49.38±5.67	45.63±4.47	48.75±3.67	52.81±3.74	1.0
D18	**48.45±6.96**	40.47±3.08	38.24±3.90	42.70±8.44	40.54±4.33	46.01±6.98	45.95±6.26	44.66±5.64	40.41±6.88	37.84±3.34	48.18±6.76	0.6
D19	62.22±3.75	63.61±0.88	64.07±4.82	63.70±5.69	64.07±0.59	64.12±0.73	62.78±3.51	62.69±3.81	62.59±4.10	64.44±1.76	**65.19±3.50**	0.8
D20	**54.48±4.14**	35.48±2.61	36.21±2.55	35.05±2.09	35.61±2.09	51.87±3.84	47.03±3.32	28.58±1.69	34.85±2.49	35.62±2.87	46.53±3.48	0.9
D21	**81.28±10.32**	53.62±6.64	59.57±9.14	56.81±7.02	59.79±8.36	72.13±6.84	68.72±3.02	50.85±5.80	53.40±6.14	55.32±7.83	70.21±2.24	0.6
D22	57.27±2.48	54.68±0.00	54.68±0.00	54.68±0.00	54.68±0.00	57.15±1.34	56.78±1.79	60.37±2.81	54.68±0.00	54.68±0.00	**64.79±0.00**	0.5
D23	**31.17±2.08**	28.54±1.84	28.59±1.84	28.49±1.60	28.15±1.94	30.83±1.45	28.68±1.41	30.10±1.81	29.46±2.58	28.15±1.94	30.63±2.49	0.7
D24	58.87±0.16	59.21±0.10	59.23±0.10	**59.27±0.09**	59.25±0.10	58.83±0.18	58.68±0.18	59.25±0.10	59.21±0.10	59.25±0.10	58.89±0.13	0.5
D25	54.90±2.31	56.03±2.80	56.99±2.73	57.14±2.89	55.67±2.63	**57.65±3.23**	56.74±3.21	57.12±3.19	56.80±2.74	**57.65±3.23**	**57.65±3.23**	0.8
D26	26.60±2.20	24.95±1.58	24.07±1.68	24.69±1.64	25.26±1.11	26.65±1.67	**26.96±1.75**	26.86±1.58	25.00±1.87	23.87±1.79	26.86±2.88	0.6
D27	52.70±0.00	**55.86±4.18**	54.65±3.14	54.00±2.74	53.35±2.06	52.70±0.00	52.70±0.00	52.70±0.00	54.00±2.74	55.30±3.36	52.70±0.00	0.5
D28	**76.30±0.00**	62.41±5.89	62.52±6.18	64.41±5.59	65.52±7.52	74.59±5.39	74.59±5.39	74.59±5.39	66.07±7.60	64.19±7.71	**76.30±0.00**	0.5
D29	61.74±0.31	62.19±0.54	62.13±0.61	61.55±2.40	**62.39±0.31**	60.45±2.22	59.81±0.44	60.06±3.00	61.35±2.33	62.19±0.54	61.35±0.64	0.8
D30	**64.24±0.80**	62.36±0.14	62.36±0.14	62.34±0.14	62.36±0.14	**64.24±0.80**	62.36±0.14	62.39±0.14	62.39±0.14	62.39±0.14	62.42±0.13	1.0
D31	77.51±12.06	68.93±0.00	68.92±0.00	68.92±0.00	68.92±0.00	77.15±7.98	74.21±10.27	77.62±8.42	68.92±0.00	68.92±0.00	**87.39±0.00**	0.5
D32	76.11±6.60	78.61±9.31	79.22±9.52	73.92±7.22	76.83±10.37	74.70±8.24	76.82±10.94	77.55±12.34	78.01±10.37	75.21±8.33	**80.60±9.84**	0.7
Avg.	57.57±2.91	55.89±3.19	55.60±3.09	55.63±3.04	55.57±3.30	57.28±3.50	56.15±3.03	56.01±3.22	55.49±3.13	55.65±2.83	**59.60±2.71**	-

Table 11
The attribute selection data on the clustering algorithm FCM ACC(%).

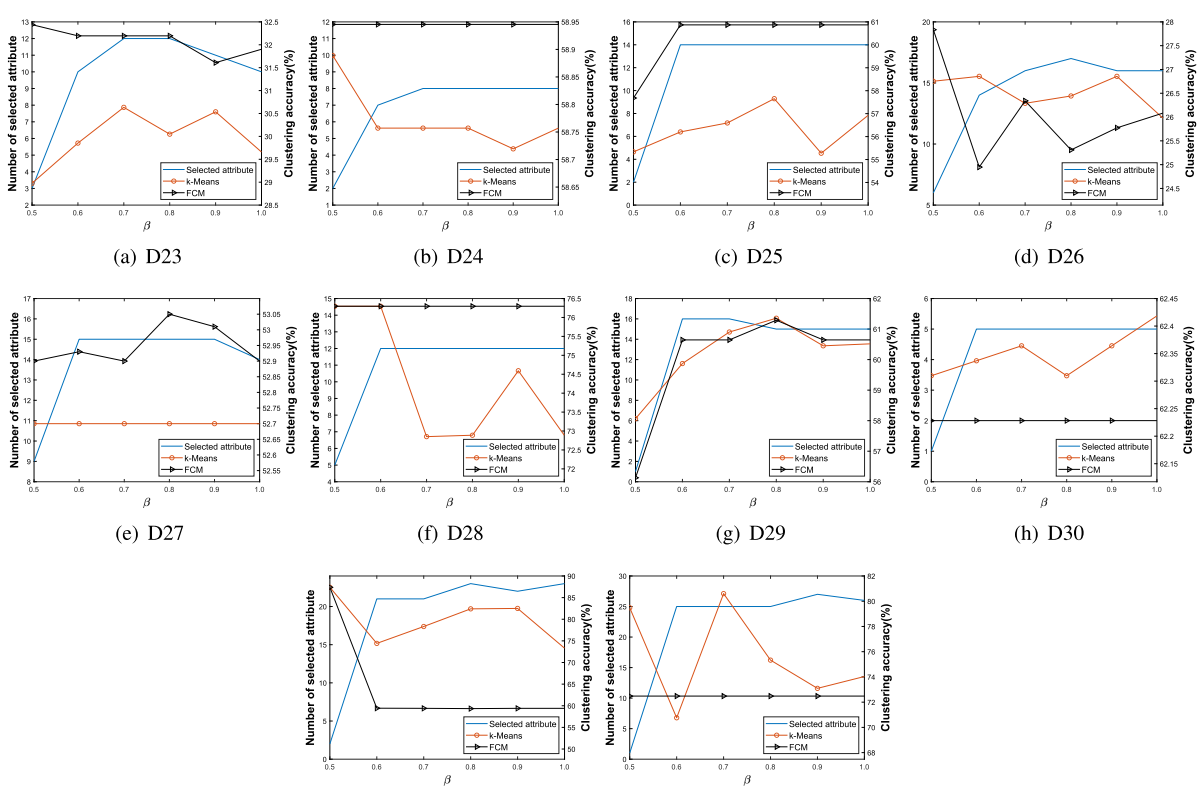
Dataset	Original attributes	LS	UFSFS	USFSM	SPEC	UFRFS	FRUAR	HKCMI	FMIUFS	UAS_MsIFMC	UAS_NAQD	β
D01	66.67±0.00	**67.06±0.00**	64.79±3.77	64.66±5.07	66.60±1.44	66.67±0.00	67.06±0.00	66.93±0.00	**67.06±0.00**	**67.06±0.00**	**67.06±0.00**	1.0
D02	51.16±2.35	48.72±1.41	49.55±1.77	48.78±1.74	48.57±1.47	**51.70±1.84**	51.67±1.79	50.21±2.26	48.75±1.69	48.33±1.51	51.13±2.31	0.9
D03	56.73±3.16	**64.30±1.68**	64.58±1.69	63.27±1.38	64.25±1.72	57.99±0.15	54.30±1.77	55.98±2.90	63.93±1.69	63.60±1.58	63.74±0.24	0.6
D04	70.94±0.00	69.80±0.00	69.80±0.00	69.80±0.00	69.80±0.00	67.52±0.00	71.23±0.00	69.52±0.00	69.80±0.00	69.80±0.00	**72.65±0.00**	0.6
D05	16.06±1.18	32.06±0.59	31.69±0.73	31.75±0.54	31.92±0.70	30.42±0.86	15.72±1.34	24.56±1.79	31.81±0.81	**32.31±0.45**	28.75±1.22	1.0
D06	67.18±0.00	69.74±0.00	69.74±0.00	69.74±0.00	69.74±0.00	**75.38±0.00**	67.69±0.00	74.87±0.00	69.74±0.00	69.74±0.00	72.82±0.00	0.9
D07	55.34±0.15	61.06±0.00	61.06±0.00	61.06±0.00	61.06±0.00	56.73±0.00	57.07±0.91	62.12±0.20	61.06±0.00	61.06±0.00	**63.46±0.00**	0.6
D08	95.28±0.00	95.28±0.00	95.28±0.00	95.28±0.00	95.28±0.00	95.28±0.00	**95.57±0.00**	95.28±0.00	95.28±0.00	95.28±0.00	**95.57±0.00**	0.7
D09	**92.79±0.00**	92.27±0.00	92.27±0.00	92.27±0.00	92.27±0.00	91.74±0.00	91.74±0.00	91.04±0.00	92.27±0.00	92.27±0.00	91.74±0.00	0.6
D10	51.44±0.00	50.32±0.02	50.32±0.01	50.32±0.00	50.31±0.01	51.44±0.00	51.48±0.00	51.60±0.00	50.31±0.01	50.32±0.02	**51.68±0.01**	1.0
D11	29.74±1.15	33.69±2.94	36.22±1.49	**36.08±2.65**	34.49±2.96	30.89±0.68	31.55±0.98	30.86±0.76	34.73±4.48	33.81±2.50	35.04±2.30	1.0
D12	58.13±0.16	59.09±0.00	59.09±0.00	59.09±0.00	59.09±0.00	58.54±2.28	58.48±2.10	**63.13±0.00**	59.09±0.00	59.09±0.00	61.62±0.00	1.0
D13	27.30±0.21	29.78±1.78	28.23±0.85	28.32±0.72	29.42±1.97	28.72±1.29	30.00±0.50	28.14±1.13	29.20±1.62	28.76±1.12	**31.73±1.77**	0.6
D14	47.14±8.56	52.61±5.03	52.69±4.73	54.99±1.99	52.38±5.96	55.06±9.79	50.30±8.08	50.72±7.21	**55.10±0.61**	51.63±5.82	54.11±2.00	0.7
D15	**39.20±4.85**	36.31±4.09	37.37±3.58	37.08±3.59	38.94±4.57	36.94±3.75	36.41±3.85	37.55±3.15	35.71±4.60	34.77±4.35	37.45±4.23	0.5
D16	52.85±3.09	51.69±0.00	51.69±0.00	51.69±0.00	51.69±0.00	52.63±3.64	51.99±2.71	55.37±1.76	51.69±0.00	51.69±0.00	**56.79±0.00**	0.6
D17	52.50±5.86	40.63±0.00	40.63±0.00	40.63±0.00	40.63±0.00	**56.88±2.47**	56.25±0.00	46.88±0.00	40.63±0.00	40.63±0.00	56.25±0.00	1.0
D18	38.51±0.00	38.51±0.00	38.51±0.00	38.51±0.00	38.51±0.00	38.51±0.00	39.05±0.28	38.51±0.00	38.51±0.00	38.58±0.21	**39.19±0.00**	0.8
D19	63.89±0.00	60.37±4.95	61.85±8.15	**66.11±6.65**	60.37±6.99	63.89±0.00	63.89±0.00	63.89±0.00	63.33±6.10	59.44±6.50	63.89±0.00	0.5
D20	24.17±0.05	26.06±1.64	27.12±1.04	26.05±1.02	25.84±1.16	23.54±0.23	24.30±0.00	31.00±0.67	26.47±0.85	26.43±1.05	**35.26±0.44**	0.5
D21	72.34±0.00	65.96±0.00	65.96±0.00	65.96±0.00	65.96±0.00	72.34±0.00	72.34±0.00	46.81±0.00	65.96±0.00	65.96±0.00	**78.72±0.00**	0.5
D22	60.41±0.36	67.04±13.03	64.57±12.76	57.15±7.82	67.04±13.03	59.93±0.00	60.71±0.12	63.30±0.00	67.04±13.03	64.57±12.76	**72.10±7.70**	0.5
D23	29.76±0.00	29.07±0.62	28.93±0.73	28.49±0.84	28.73±0.87	29.76±0.00	31.61±1.85	28.05±0.77	28.59±0.90	28.73±0.87	**32.44±0.77**	0.5
D24	**58.95±0.00**	**58.95±0.00**	**58.95±0.00**	**58.95±0.00**	**58.95±0.00**	**58.95±0.00**	**58.95±0.00**	**58.95±0.00**	**58.95±0.00**	**58.95±0.00**	**58.95±0.00**	0.5
D25	60.43±0.00	60.43±0.00	60.43±0.00	60.43±0.00	60.43±0.00	60.43±0.00	**60.87±0.00**	58.99±0.00	60.43±0.00	60.43±0.00	**60.87±0.00**	0.6
D26	25.41±0.25	24.85±1.49	25.26±1.63	24.64±1.49	25.26±1.63	25.24±0.16	26.24±0.82	**28.51±0.42**	25.46±1.54	24.59±1.88	27.84±0.00	0.5
D27	53.00±0.00	53.00±0.00	53.00±0.00	53.01±0.03	53.00±0.00	53.00±0.00	52.97±0.08	52.80±0.00	53.00±0.00	53.00±0.00	**53.05±0.12**	0.8
D28	**76.30±0.00**	58.52±0.00	58.52±0.00	58.52±0.00	58.52±0.00	76.30±0.00	**76.30±0.00**	**76.30±0.00**	58.52±0.00	58.52±0.00	**76.30±0.00**	0.5
D29	**61.94**±0.00	60.65±0.00	60.65±0.00	60.65±0.00	60.65±0.00	**61.94±0.00**	60.65±0.00	59.35±0.00	60.65±0.00	60.65±0.00	61.29±0.00	0.8
D30	**63.86±0.00**	57.17±6.30	62.39±6.30	61.09±6.74	61.09±6.74	**63.86±0.00**	62.23±0.00	62.23±0.00	59.78±6.87	58.48±6.74	62.23±0.00	0.5
D31	59.28±0.00	68.92±0.00	68.92±0.00	68.92±0.00	68.92±0.00	59.41±0.00	59.41±0.00	68.92±0.00	68.92±0.00	68.92±0.00	**87.39±0.00**	0.5
D32	**72.48±0.00**	53.15±0.00	53.15±0.00	53.15±0.00	53.15±0.00	**72.48±0.00**	**72.48±0.00**	**72.48±0.00**	53.15±0.00	53.15±0.00	**72.48±0.00**	0.5
Avg.	54.72±0.98	54.28±1.42	54.48±1.54	54.26±1.32	54.46±1.60	55.78±0.85	55.02±0.85	55.15±0.72	54.53±1.40	54.08±1.48	**58.55±0.72**	-

Table 12
ACC ordinal values based on the clustering algorithm k-Means.

Dataset	LS	UFSFS	USFSM	SPEC	UFRFS	FRUAR	HKCMI	FMIUFS	UAS_MsIFMC	UAS_NAQD
D01	3.0	5.0	8.0	9.0	4.0	6.0	7.0	2.0	1.0	10.0
D02	3.0	5.0	4.0	1.0	6.0	2.0	8.0	9.0	7.0	10.0
D03	7.0	5.0	4.0	9.0	2.0	1.0	3.0	6.0	10.0	8.0
D04	7.5	4.0	5.0	7.5	1.0	2.0	9.0	6.0	3.0	10.0
D05	4.0	3.0	6.0	5.0	1.0	8.0	9.0	7.0	2.0	10.0
D06	5.0	7.0	8.0	4.0	9.0	2.0	1.0	3.0	6.0	10.0
D07	7.0	8.0	5.0	9.0	3.0	2.0	1.0	4.0	6.0	10.0
D08	5.5	5.5	5.5	5.5	3.0	1.0	9.0	9.0	9.0	2.0
D09	4.0	3.0	5.0	1.0	9.0	2.0	10.0	6.0	7.0	8.0
D10	10.0	1.0	4.0	7.0	8.0	3.0	2.0	6.0	5.0	9.0
D11	5.0	1.0	2.0	3.0	7.0	9.0	8.0	6.0	4.0	10.0
D12	7.0	8.0	9.0	5.0	6.0	1.0	4.0	2.0	3.0	10.0
D13	8.5	5.0	10.0	8.5	3.0	1.0	2.0	6.5	6.5	4.0
D14	4.0	8.0	3.0	9.0	6.0	1.0	7.0	2.0	5.0	10.0
D15	8.0	2.0	10.0	7.0	3.0	1.0	5.0	6.0	4.0	9.0
D16	5.5	5.5	5.5	5.5	8.0	1.0	9.0	2.5	2.5	10.0
D17	4.5	3.0	1.0	4.5	9.0	10.0	7.0	2.0	6.0	8.0
D18	4.0	2.0	6.0	5.0	9.0	8.0	7.0	3.0	1.0	10.0
D19	4.0	6.5	5.0	6.5	8.0	3.0	2.0	1.0	9.0	10.0
D20	4.0	7.0	3.0	5.0	10.0	9.0	1.0	2.0	6.0	8.0
D21	3.0	6.0	5.0	7.0	10.0	8.0	1.0	2.0	4.0	9.0
D22	2.5	2.5	2.5	2.5	8.0	7.0	9.0	5.5	5.5	10.0
D23	3.0	4.0	2.0	1.0	10.0	6.0	8.0	7.0	5.0	9.0
D24	5.0	6.0	10.0	7.0	2.0	1.0	8.5	4.0	8.5	3.0
D25	2.0	5.0	7.0	1.0	9.5	3.0	6.0	4.0	8.0	9.5
D26	4.0	2.0	3.0	6.0	7.0	10.0	8.0	5.0	1.0	9.0
D27	10.0	8.0	6.5	5.0	2.5	2.5	2.5	6.5	9.0	2.5
D28	1.0	2.0	4.0	5.0	9.0	7.5	7.5	6.0	3.0	10.0
D29	9.0	7.0	6.0	10.0	3.0	1.0	2.0	5.0	8.0	4.0
D30	3.0	3.0	1.0	3.0	10.0	5.0	7.0	7.0	7.0	9.0
D31	2.5	2.5	2.5	2.5	8.0	7.0	9.0	5.5	5.5	10.0
D32	8.0	9.0	1.0	5.0	2.0	4.0	6.0	7.0	3.0	10.0
Avg.	5.1	4.7	5.0	5.4	6.1	4.2	5.8	4.9	5.3	8.5

Table 13
ACC ordinal values based on the clustering algorithm FCM.

Dataset	LS	UFSFS	USFSM	SPEC	UFRFS	FRUAR	HKCMI	FMIUFS	UAS_MsIFMC	UAS_NAQD
D01	8.0	2.0	1.0	3.0	4.0	8.0	5.0	8.0	8.0	8.0
D02	3.0	6.0	5.0	2.0	10.0	9.0	7.0	4.0	1.0	8.0
D03	9.0	10.0	4.0	8.0	3.0	1.0	2.0	7.0	5.0	6.0
D04	5.5	5.5	5.5	5.5	1.0	9.0	2.0	5.5	5.5	10.0
D05	9.0	5.0	6.0	8.0	4.0	1.0	2.0	7.0	10.0	3.0
D06	4.5	4.5	4.5	4.5	10.0	1.0	9.0	4.5	4.5	8.0
D07	5.5	5.5	5.5	5.5	1.0	2.0	9.0	5.5	5.5	10.0
D08	4.5	4.5	4.5	4.5	4.5	9.0	4.5	4.5	4.5	9.0
D09	7.5	7.5	7.5	7.5	2.0	2.0	1.0	7.5	7.5	2.0
D10	4.0	4.0	4.0	2.0	7.0	8.0	9.0	1.0	6.0	10.0
D11	4.0	10.0	9.0	6.0	2.0	3.0	1.0	7.0	5.0	8.0
D12	5.5	5.5	5.5	5.5	2.0	1.0	10.0	5.5	5.5	9.0
D13	8.0	2.0	3.0	7.0	4.0	9.0	1.0	6.0	5.0	10.0
D14	5.0	6.0	8.0	4.0	9.0	1.0	2.0	10.0	3.0	7.0
D15	3.0	7.0	6.0	10.0	5.0	4.0	9.0	2.0	1.0	8.0
D16	3.5	3.5	3.5	3.5	8.0	7.0	9.0	3.5	3.5	10.0
D17	3.5	3.5	3.5	3.5	10.0	8.0	7.0	3.5	3.5	8.0
D18	7.0	7.0	7.0	7.0	7.0	9.0	7.0	7.0	8.0	10.0
D19	2.5	4.0	10.0	2.5	7.5	7.5	7.5	5.0	1.0	7.5
D20	5.0	8.0	4.0	3.0	1.0	2.0	9.0	7.0	6.0	10.0
D21	4.5	4.5	4.5	4.5	8.5	8.5	1.0	4.5	4.5	10.0
D22	8.0	5.5	1.0	8.0	2.0	3.0	4.0	8.0	5.5	10.0
D23	7.0	6.0	2.0	4.5	8.0	9.0	1.0	3.0	4.5	10.0
D24	5.5	5.5	5.5	5.5	5.5	5.5	5.5	5.5	5.5	5.5
D25	5.0	5.0	5.0	5.0	5.0	9.5	1.0	5.0	5.0	9.5
D26	3.0	4.5	2.0	4.5	7.5	7.5	10.0	6.0	1.0	9.0
D27	5.5	5.5	9.0	5.5	5.5	2.0	1.0	5.5	5.5	10.0
D28	3.5	3.5	3.5	3.5	8.5	8.5	8.5	3.5	3.5	8.5
D29	5.0	5.0	5.0	5.0	10.0	5.0	1.0	5.0	5.0	9.0
D30	1.0	9.0	4.5	4.5	10.0	7.0	7.0	3.0	2.0	7.0
D31	6.0	6.0	6.0	6.0	1.5	1.5	6.0	6.0	6.0	10.0
D32	3.5	3.5	3.5	3.5	8.5	8.5	8.5	3.5	3.5	8.5
Avg.	5.2	5.5	5.0	5.1	5.7	5.5	5.2	5.3	4.7	8.4



(a) D23 (b) D24 (c) D25 (d) D26

(e) D27 (f) D28 (g) D29 (h) D30

(i) D31 (j) D32

Fig. 6. The number of selected attributes and ACC on hybrid attribute datasets using different β values.

the attribute subsequence with the same number of attributes selected by UAS_NAQD. Finally, we choose the attribute subsequence with the highest ACC as the experimental result. The parameter β of UAS_NAQD is used to adjust the partition of the positive region. The parameter β has a big impact on UAS_NAQD. In general, the different values of β will lead to different results. Therefore, we need to adjust β from 0.5 to 1.0 with a step size of 0.1 to obtain the optimal result. Many experimental results show that for numerical and hybrid attribute datasets, UAS_UAQD usually has difficulty achieving the stopping condition that the significance of all remaining attributes is 0. Therefore, this paper sets the significance of the stopping condition to 0.0001. At the same time, this paper sets the lower limit of the increase rate of the average VPNCCR $\mathcal{I}nc_ctrl = 0.0001$. If $\mathcal{I}nc(c_k, \mathcal{P}, c_s) = 0$, then $\mathcal{I}nc(c_k, \mathcal{P}, c_s) = \mathcal{I}nc_ctrl$. Finally, an optimal subset of attributes is selected for each dataset. The labels of all datasets will be removed during attribute selection.

5.3. Experimental results

In this section, we will analyze the results of the clustering experiments. The comparative results of the clustering experiments are shown in Tables 8–11. Tables 8 and 9 give the optimal number of attributes based on the clustering algorithms k-Means and FCM, respectively. Where bold indicates that the results obtain by the corresponding UAS algorithm is the original attribute set.

From Tables 8 and 9, we can find that all UAS algorithms can remove some irrelevant or redundant attributes on some datasets in most cases. However, from Tables 8 and 9, we can find that the algorithm UFRFS can not remove any attributes on some datasets. For example, datasets D01, D02, D08, D14, D15, D19, D25, and D28. The number of attributes selected by UAS_NAQD is smaller than the original number of attributes on all datasets and is significantly smaller on some datasets. For example, for dataset D05, UAS_NAQD selects the number of attributes as 9 and 9, respectively; for dataset D07, UAS_NAQD selects the number of attributes as 10 and 9, respectively; for dataset D13, UAS_NAQD selects the number of attributes as 4 and 15, respectively; for dataset D17, UAS_NAQD selects the number of attributes as 5 and 5, respectively; for dataset D24, UAS_NAQD selects the number of attributes as 2 and 7, respectively; for dataset D31, UAS_NAQD selects the number of attributes as 2 and 7, respectively. In addition, UAS_NAQD is also smaller than most other UAS algorithms for the mean value, which indicates that UAS_NAQD can remove irrelevant and redundant attributes more effectively.

Tables 10 and 11 present the comparison of ACC based on k-Means and FCM, respectively. Where bold indicates the optimal ACC of different UAS algorithms. From Tables 10 and 11, we can see that UAS_NAQD can maintain or improve ACC on most datasets. The detailed analyses are as follows.

(1) Compared with the other 9 UAS algorithms, UAS_NAQD achieves the best ACC in 34 cases. For algorithms LS, UFSFS, USFSM, SPEC, UFRFS, FRUAR, HKCMI, FMIUFS, and UAS_MsIFMC are only 6, 4, 7, 3, 10, 8, 6, 4, and 6, respectively. In addition, the ACC of UAS_NAQD is 53, 53, 52, 53, 43, 48, 50, 53, and 53 higher than that of algorithms LS, UFSFS, USFSM, SPEC, UFRFS, FRUAR, HKCMI, FMIUFS, and UAS_MsIFMC, respectively.
(2) From the average value of ACC, UAS_NAQD achieves the higher average value.
(3) From the standard deviation of ACC, UAS_NAQD achieves the lower average value.

In summary, UAS_NAQD achieves the optimal ACC in most cases. Therefore, UAS_NAQD applies to uncertain hybrid data to attribute selection for clustering tasks.

Previously, we have analyzed the results of clustering experiments. Finally, we have the following findings based on the comprehensive analysis of the above experimental results.

(1) In most cases, the greater the number of conditional attributes, the more efficient the UAS will be. The more labels, the lower the ACC.
(2) For datasets containing nominal attributes, replacing nominal attributes with different integer values also affects the clustering performance of algorithms LS, UFSFS, and UFRFS. Although the algorithm USFSM applies to hybrid attribute data, it does not obtain better results. This is probably because it can not handle uncertain data.
(3) UAS_NAQD requires neither substitution nor discretization and retains more of the true information of the data. At the same time, UAS_NAQD takes advantage of VPNRS, which can effectively handle uncertain hybrid data. Therefore, UAS_NAQD has the better clustering performance.

5.4. Statistical test

This section uses the method of Refs. (Hu et al., 2025; Wan et al., 2025) for hypothesis testing (Friedman's test and Nemenyi's post-hoc test). We will utilize it to evaluate the statistical significance of UAS algorithms. Before using Friedman's test, the ACC values of each UAS algorithm on all the datasets are first ranked in descending order. The sequence number is assigned to $(1, 2, 3, \cdots)$. If the ACC values of several UAS algorithms are the same, we divide their order values equally, which are shown in Tables 12 and 13.

Next, we will use Friedman's test to determine whether these UAS algorithms have the same performance. Suppose that $\mathcal{M}$ UAS algorithms are compared on $\mathcal{N}$ datasets. r_i represents the average ordinal value of the ith UAS algorithm. The Friedman's test is computed as follows.

$$\tau_{\chi^2} = \frac{12\mathcal{N}}{\mathcal{M}(\mathcal{M}+1)}\left(\sum_{i=1}^{\mathcal{M}} r_i^2 - \frac{\mathcal{M}(\mathcal{M}+1)^2}{4}\right), \tau_{\mathcal{F}} = \frac{(\mathcal{N}-1)\tau_{\chi^2}}{\mathcal{N}(\mathcal{M}-1)-\tau_{\chi^2}}. \tag{23}$$

Where $\tau_{\mathcal{F}}$ obeys the $\mathcal{F}$ distribution with $(\mathcal{M}-1)$ and $(\mathcal{M}-1)(\mathcal{N}-1)$ degrees of freedom. If the original supposition that "all UAS algorithms have the same performance" is rejected. It shows that the UAS algorithms have significant differences in performance. At this point, Nemenyi's post-hoc test is usually used to further distinguish these UAS algorithms based on the clustering algorithms. The Critical Difference (CD) of the average ordinal value, which is computed as

$$CD_\alpha = q_\alpha\sqrt{\frac{\mathcal{M}(\mathcal{M}+1)}{6\mathcal{N}}}, \tag{24}$$

where q_α is the critical value of the Tukey distribution, we can find it in the Ref. Li et al. (2024b).

Given a level of significance α, we can obtain the corresponding CD_α. By calculating the difference of the average order value of any two UAS algorithms on the same clustering algorithm. If the difference between them is greater than CD_α, it shows that there are significant differences in the performance of the two UAS algorithms on the clustering algorithms. For instance, in Table 12, the difference between the average ordinal values of algorithms UAS_NAQD and UAS_MsIFMC is 3.2, which is greater than CD_α; in Table 13, the average ordinal difference between algorithms UAS_NAQD and UAS_MsIFMC is 3.7, which is larger than CD_α. This shows that there are significant differences in the performance of algorithms UAS_NAQD and UAS_MsIFMC on the clustering algorithms. However, the above methods cannot intuitively describe the significant differences between any two UAS algorithms. To this end, we present the above experimental results using Nemenyi's test figure (Zhang et al., 2024). From Tables 10 and 11, we can get $\mathcal{M} = 10$ and $\mathcal{N} = 32$, the τ_F distribution has 9 and 278 degrees of freedom. According to Friedman's test, the clustering algorithms k-Means and FCM have τ_F and critical difference ($\alpha = 0.1$), which are shown in Table 14. From Table 14, we can see that the value of τ_F on the clustering algorithms is greater than the critical value of 1.6541. This shows that the performance of all UAS algorithms is significantly different from the clustering algorithms. At this point, we need to further distinguish the performance of these UAS algorithms using Nemenyi's post-hoc test. For $\alpha = 0.1$, we can get the corresponding $CD_{0.1} = 2.2102$.

Table 14
The τ_F values of different clustering algorithms.

Clustering algorithm	τ_F	Critical value ($\alpha = 0.1$)	$CD_{\alpha=0.1}$
k-Means	5.4805	1.6541	2.2102
FCM	4.5155		

Nemenyi's test figure on clustering algorithms, which is shown in Fig. 3. UAS_NAQD is significantly different from most UAS algorithms. For instance, from Fig. 3(a) and Fig. 3(b), we can see that UAS_NAQD does not have any horizontal line segments connected with other UAS algorithms. This shows that UAS_NAQD is statistically significantly different from other UAS algorithms w.r.t. the clustering algorithms. In addition, UAS_NAQD has a relatively large average ordinal value on the clustering algorithms, which also shows that UAS_NAQD proposed in this paper has good effectiveness.

5.5. Parameter sensitivity analysis

The parameter β plays a very important role in UAS_NAQD, which is used to control the neighborhood granules of data analysis. For UAS_NAQD, we can get different subsets of attributes at each level of granularity by β. On this basis, we can obtain a subset of attributes describing the recognition problem of different clustering algorithms. Figs. 4, 5, and 6 show the numerical, nominal, and hybrid attribute datasets, which describe the number of selected attributes and ACC curves with different β values. Next, we will analyze both the number of selected attributes and ACC.

From Figs. 4 - 6, we can see that on some datasets, as β increases, the number of selected attributes increases steeply and then remains relatively state, for instance, datasets D01, D02, D07, D08, D09, D11, D12, D16, D19, D20, D22, D24, D25, D26, D27, D28, D29, D30, D31, and D32, where hybrid attribute datasets basically present this situation. For datasets D03, D05, D13, and D18, the number of selected attributes increases gradually as β increases and then remains relatively steady. For datasets D04, D06, and D17, the number of selected attributes increases steeply to the maximum value as β increases and then decreases gradually, finally remaining relatively steady. For the nominal attribute datasets D14 and D15, the number of selected attributes remains relatively steady as β increases.

In terms of ACC, from Figs. 4 - 6, we can see that for most datasets, the optimal value can be obtained under multiple parameter β values. According to Figs. 4 - 6, we can choose an optimal parameter β for each dataset. The parameter β ranges from 0.5 to 1.0 with a step size of 0.1. From the figures, we can find that when $\beta < 0.7$, UAS_NAQD identifies fewer attributes and the corresponding ACC is not high; when $\delta > 0.9$, the number of attributes found by UAS_NAQD shows a decreasing trend, and the corresponding ACC also shows a decreasing trend. The parameter in these regions is not easy to use, and $\beta \in [0.7, 0.9]$ is relatively more desirable. Therefore, UAS_NAQD is feasible for attribute selection in clustering algorithms. From the above analysis, we can see that β has a certain effect on the experimental performance of UAS_NAQD. Therefore, it is necessary to adjust β to obtain the optimal result. With the right β value, UAS_NAQD can obtain a better result in most cases. In summary, UAS_NAQD is feasible and effective.

6. Conclusion

In this paper, we propose an unsupervised hybrid attribute selection method based on neighborhood attribute quality degree. The method not only takes advantage of the VPND that can effectively handle uncertain data but also considers the increased rate of the average VPNCCR based on all single-condition attribute subsets. In addition, we use the hybrid neighborhood similarity measure to compute the neighborhood relation. This makes the proposed method apply to numerical, nominal, and hybrid attribute data. The method avoids the discretization of numerical attribute data, thus reducing data processing time and information loss. For this method, we have designed the corresponding algorithm UAS_NAQD. The 32 datasets from publicly available databases were selected to compare and analyze UAS_NAQD with existing UAS algorithms. Experimental results show that UAS_NAQD can identify an optimal subset of attributes to maintain or improve the performance of the clustering algorithms. It applies to uncertain hybrid data. In addition, statistical test results show that UAS_NAQD is statistically significantly different from most existing UAS algorithms.

Although UAS_NAQD has yielded promising results, it still exhibits certain limitations, thereby opening pathways for future research. First, the model relies on several key parameters, most notably the variable precision threshold parameter β. Therefore, an important research direction for future work is to develop an adaptive mechanism that automatically adjusts these parameters. Second, the time complexity of this algorithm is high. The granular-ball computing paradigm shifts the model processing unit from samples to granules, which can not only enhance computational efficiency but also effectively leverage the inherent multi-granularity information within the data. In the future, we can study the behavior of unsupervised allocation of hybrid attribute data based on the adaptive neighborhood granular-ball

CRediT authorship contribution statement

Yi Li: Conceptualization, Methodology, Writing – original draft; **Shenhong Lei:** Software, Investigation; **Xinyu Su:** Conceptualization, Methodology; **Zhong Yuan:** Supervision, Writing – review & editing; **Ji Li:** Writing – review & editing; **Xingqiang Tan:** Funding acquisition, Supervision, Writing – review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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